

Optimal Control for Space Systems

EN.530.626 (Fall 2026)

Lecture 12

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Course Assistant 1: Marvat Johri

Course Assistant 2: Patrick Schwartz

Class review

- So far: smooth constrained optimization

$$\begin{array}{ll} \min_x f(x) & f \in \mathcal{C}^2 \\ \text{subject to: } & h_i(x) = 0, \quad i = 1, \dots, m \quad h_i \in \mathcal{C}^2 \\ & g_i(x) \leq 0, \quad i = 0, \dots, p \quad g_i \in \mathcal{C}^2 \end{array}$$

- Allows us to leverage Newton method-style approaches for solving problem

Class review

Quadratic programs

- One particular formulation we have returned to has been *convex* quadratic programs

$$\begin{array}{ll} \min_x & \frac{1}{2}x^T P x + p^T x & P \in \mathbb{S}_+^n \\ \text{subject to:} & Ax = 0 & A \in \mathbb{R}^{m \times n} \\ & Gx \leq h & G \in \mathbb{R}^{p \times n} \end{array}$$

- Applied primal-dual interior point methods to find solutions for these

Recall

- How to determine if $x^\star$ satisfying N.C.O. is a local minimizer?
- How many local minimizers are there?
- Is a local minimizer the global one?

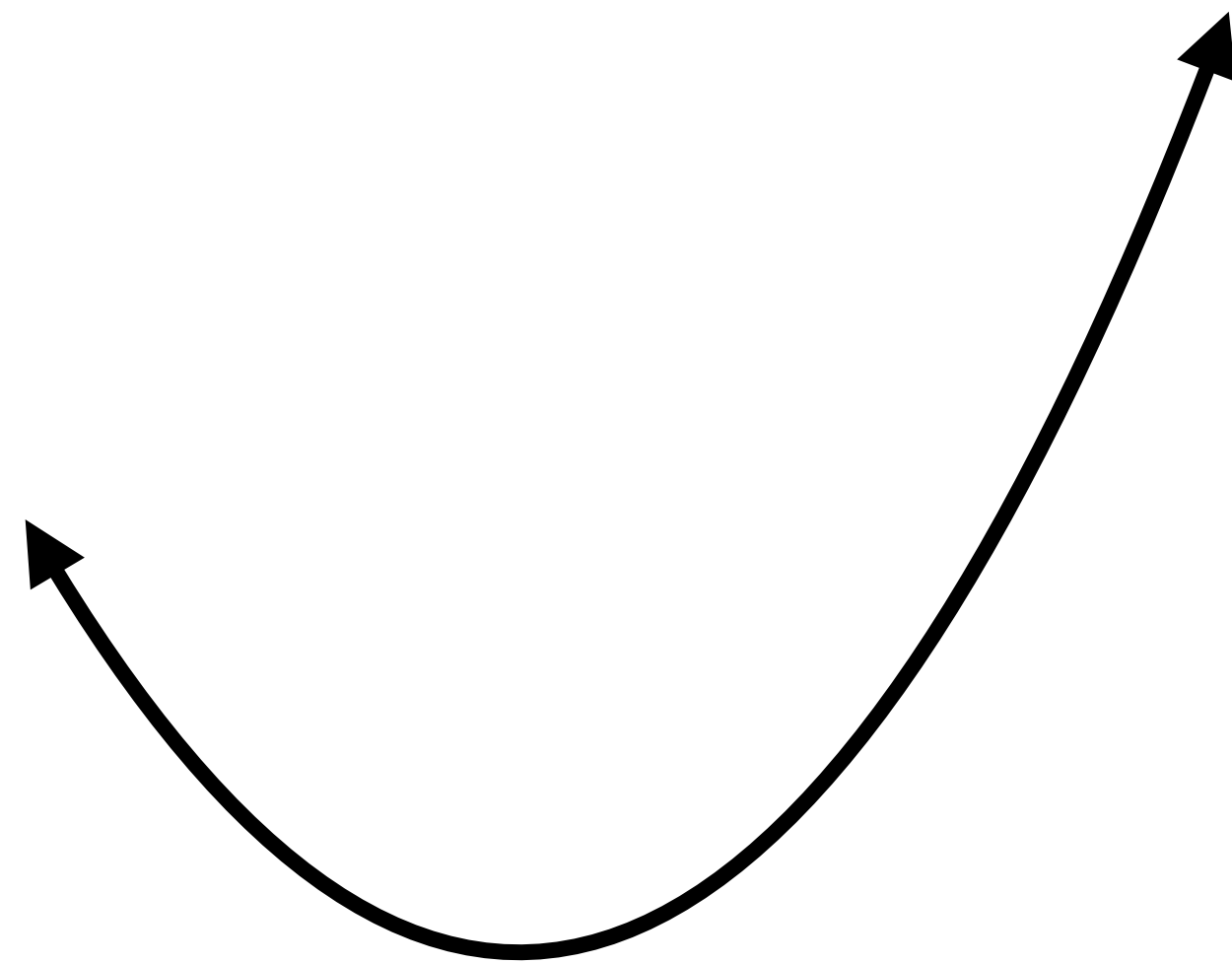
What's next?

In this class, we'll broadly study the following categories of optimization problems:

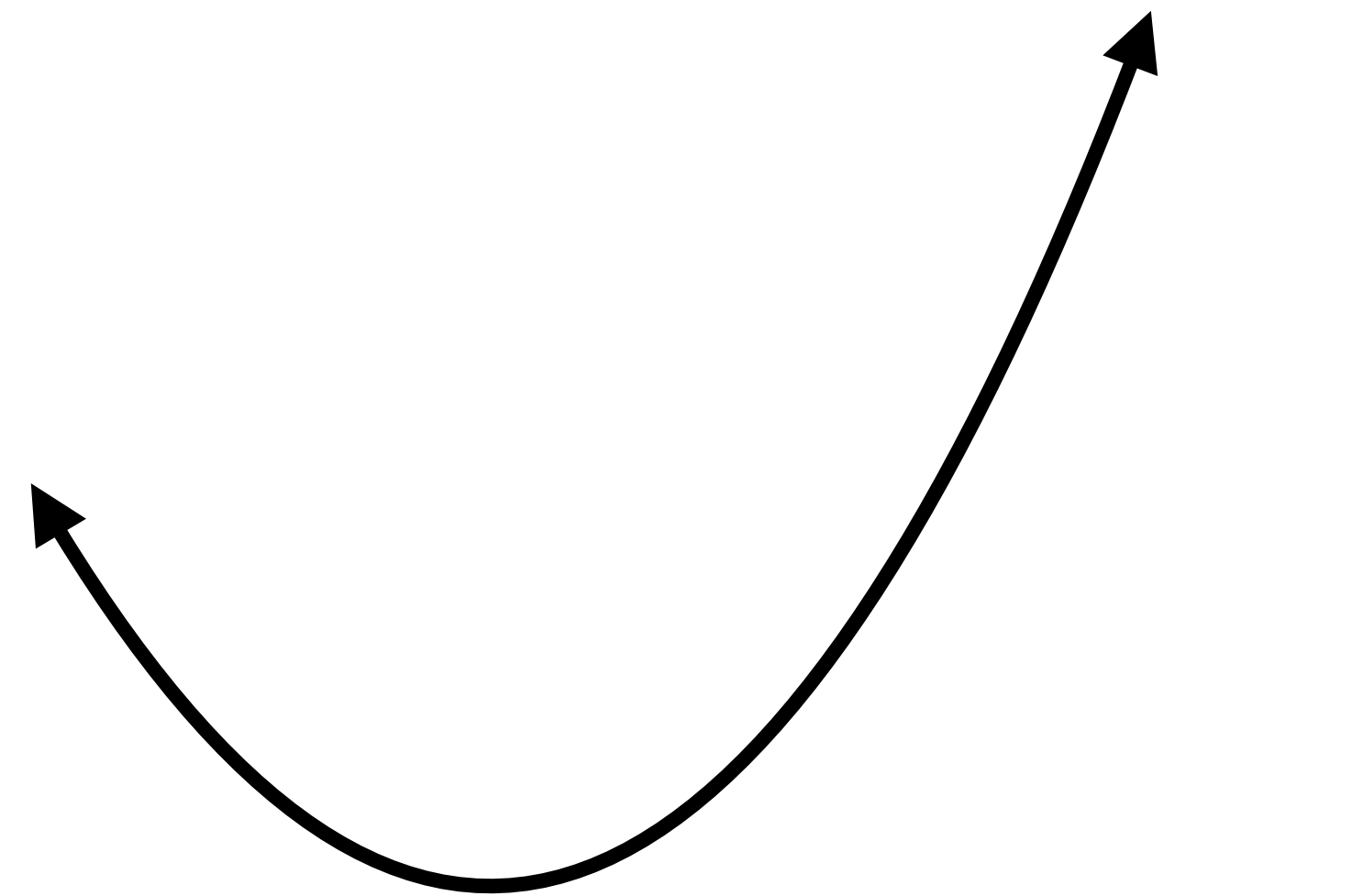
- Unconstrained vs. constrained
- Smooth vs. non-smooth
- Continuous vs. discrete
- Deterministic vs. stochastic
- Convex vs. non-convex

What's next?

Unconstrained vs. constrained



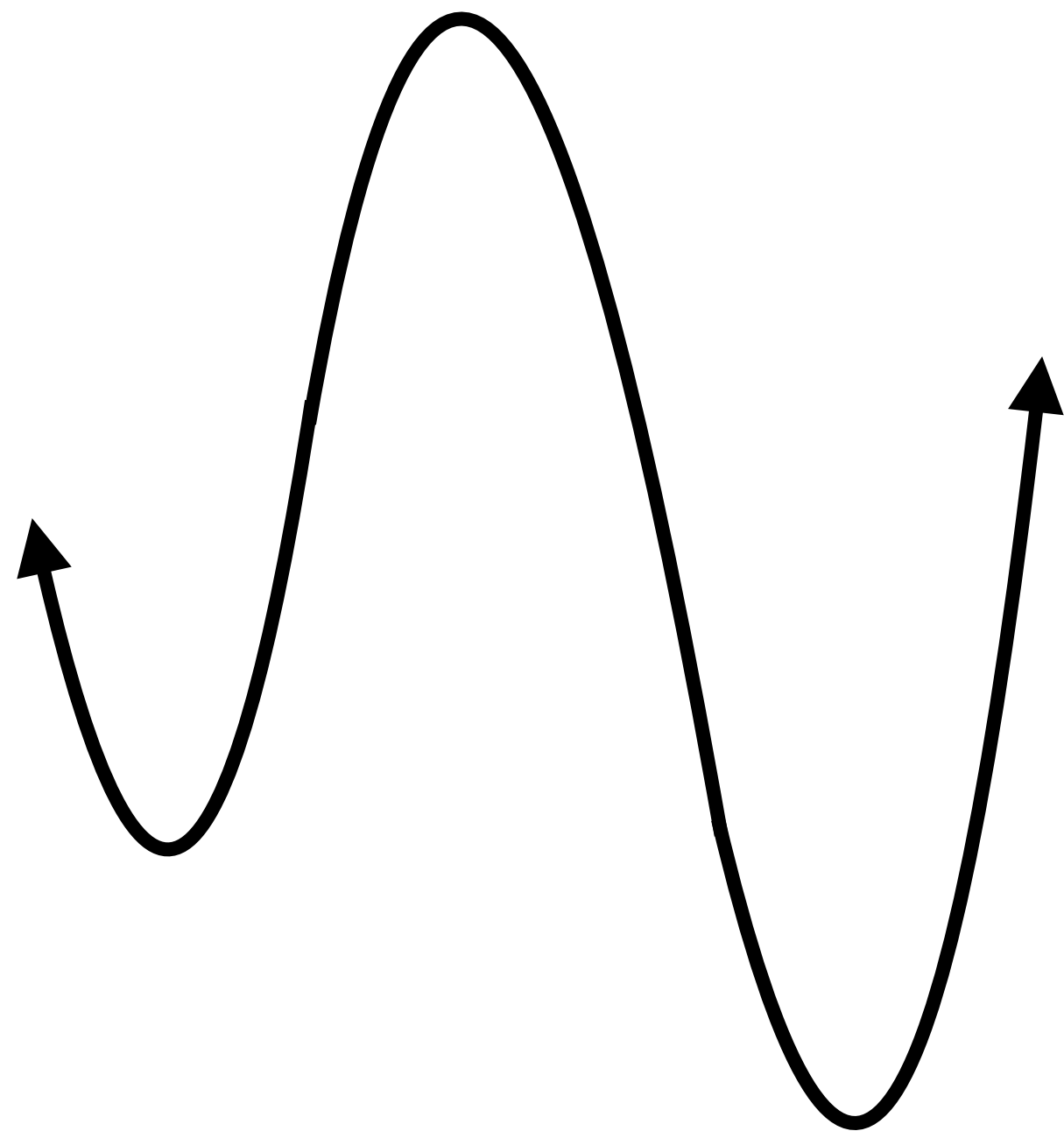
Unconstrained



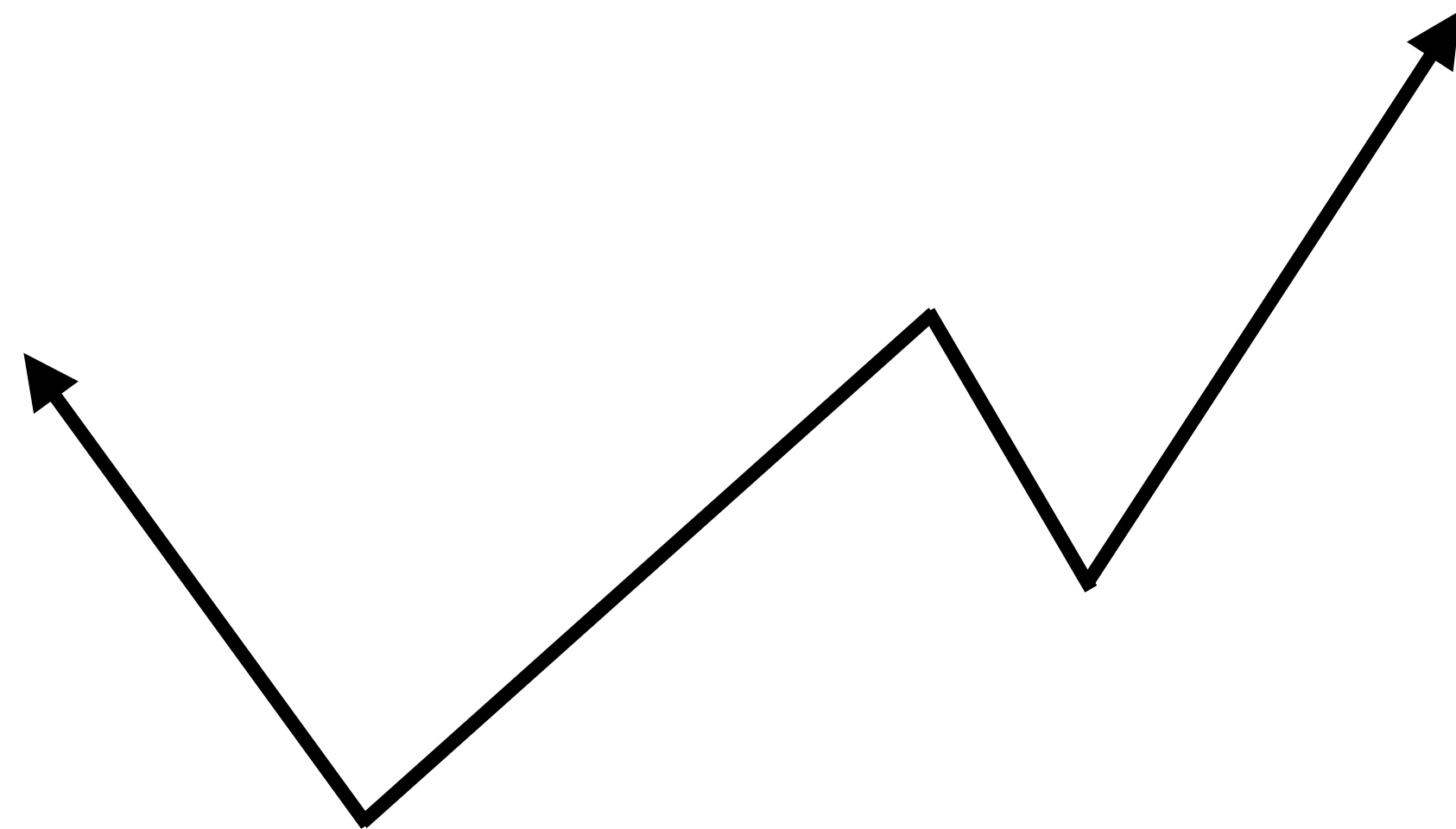
Constrained

What's next?

Smooth vs. non-smooth



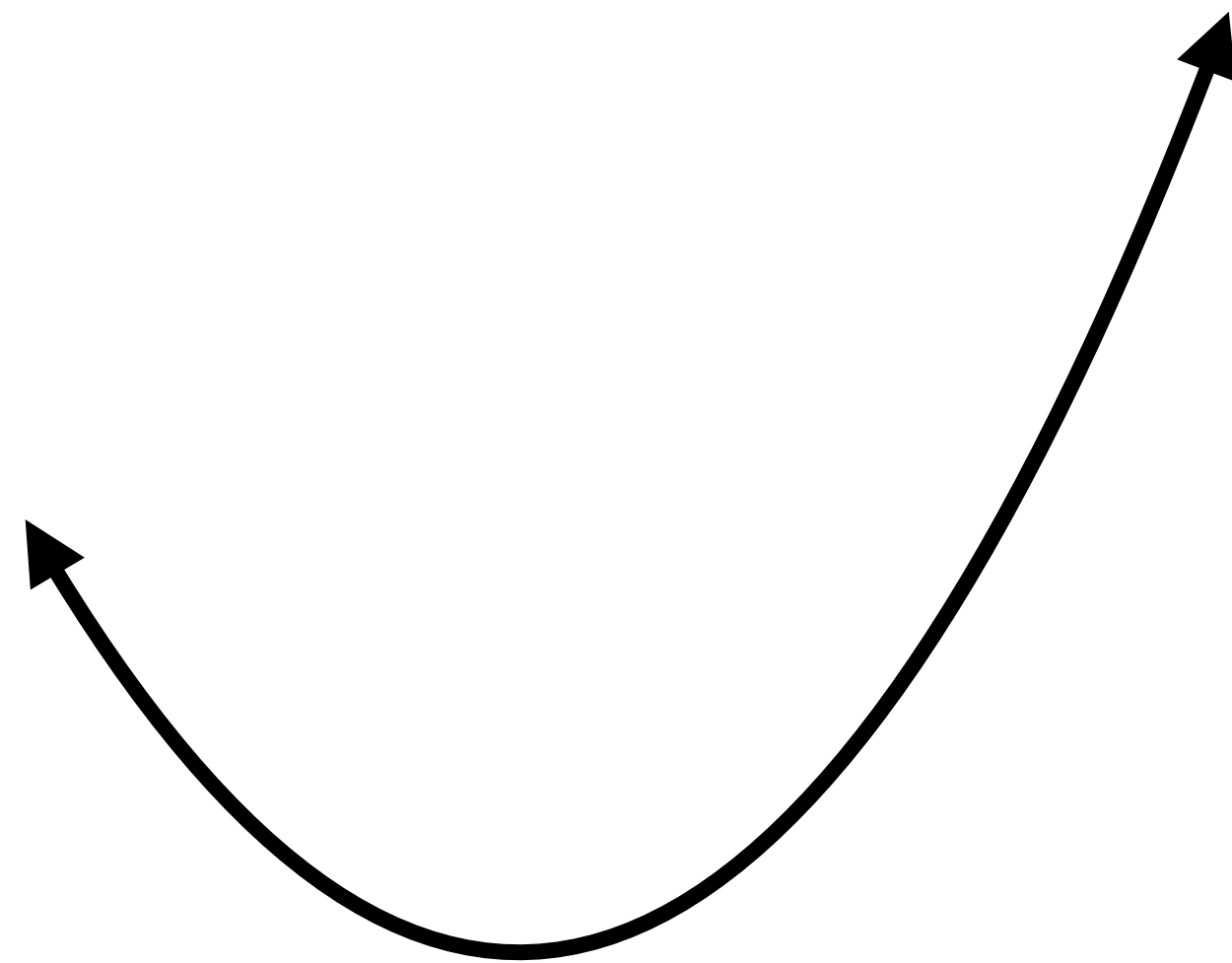
Smooth



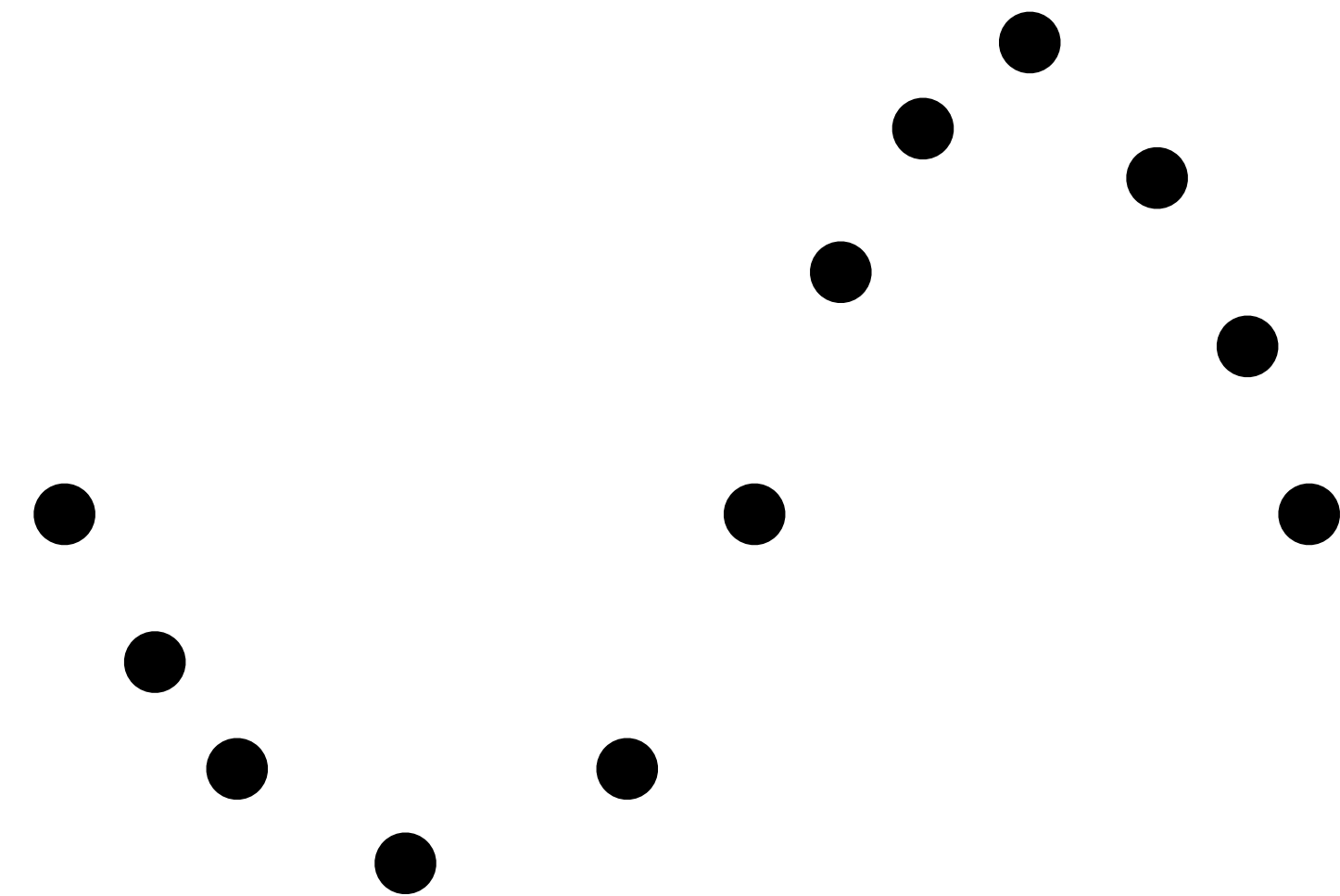
Non-smooth

What's next?

Continuous vs. discrete



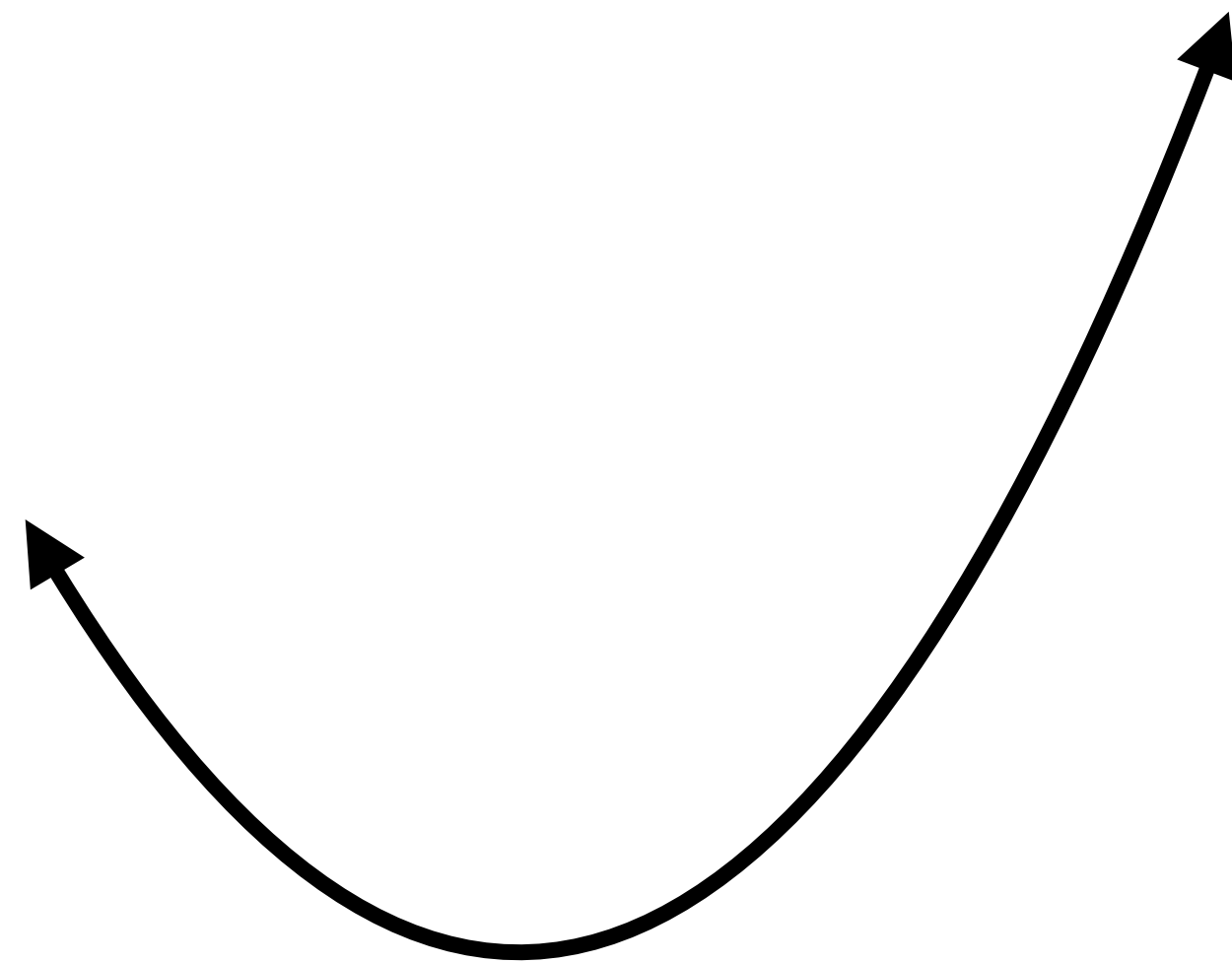
Continuous



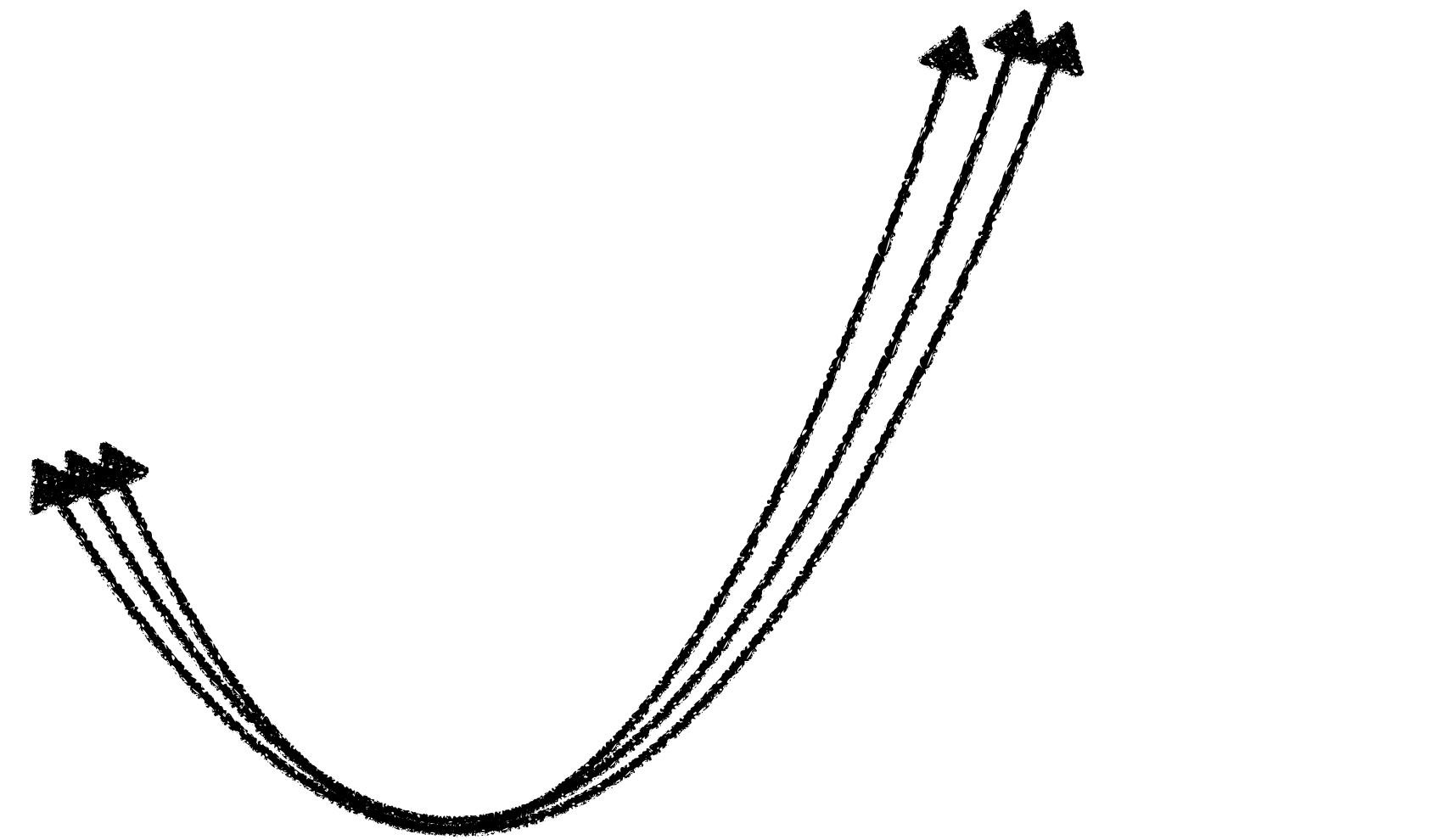
Discrete

What's next?

Deterministic vs. stochastic



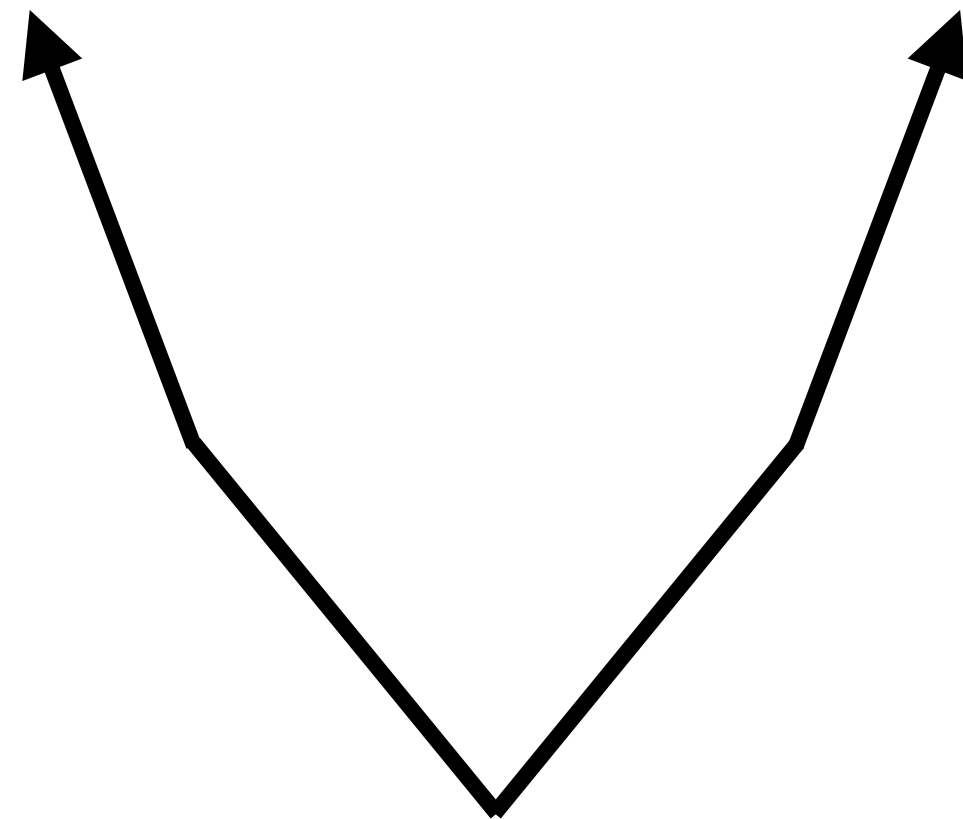
Deterministic



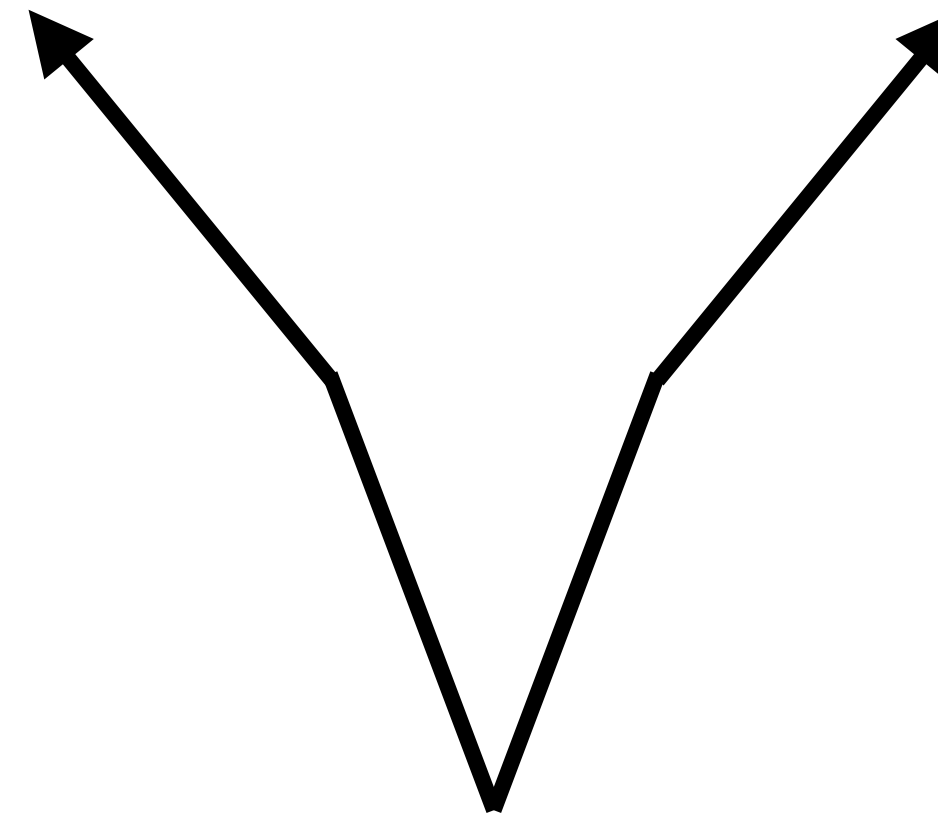
Stochastic

What's next?

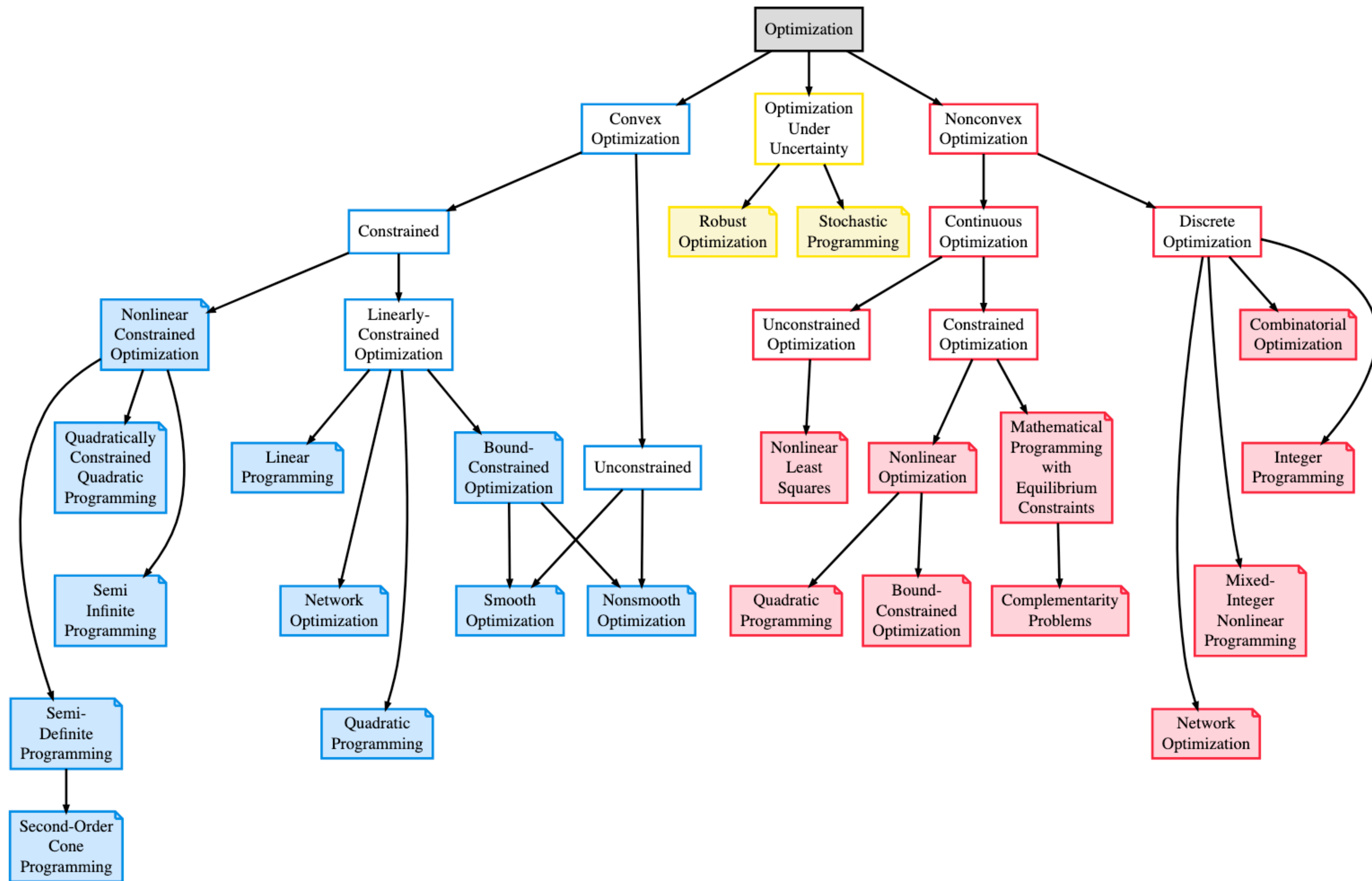
Convex vs. non-convex

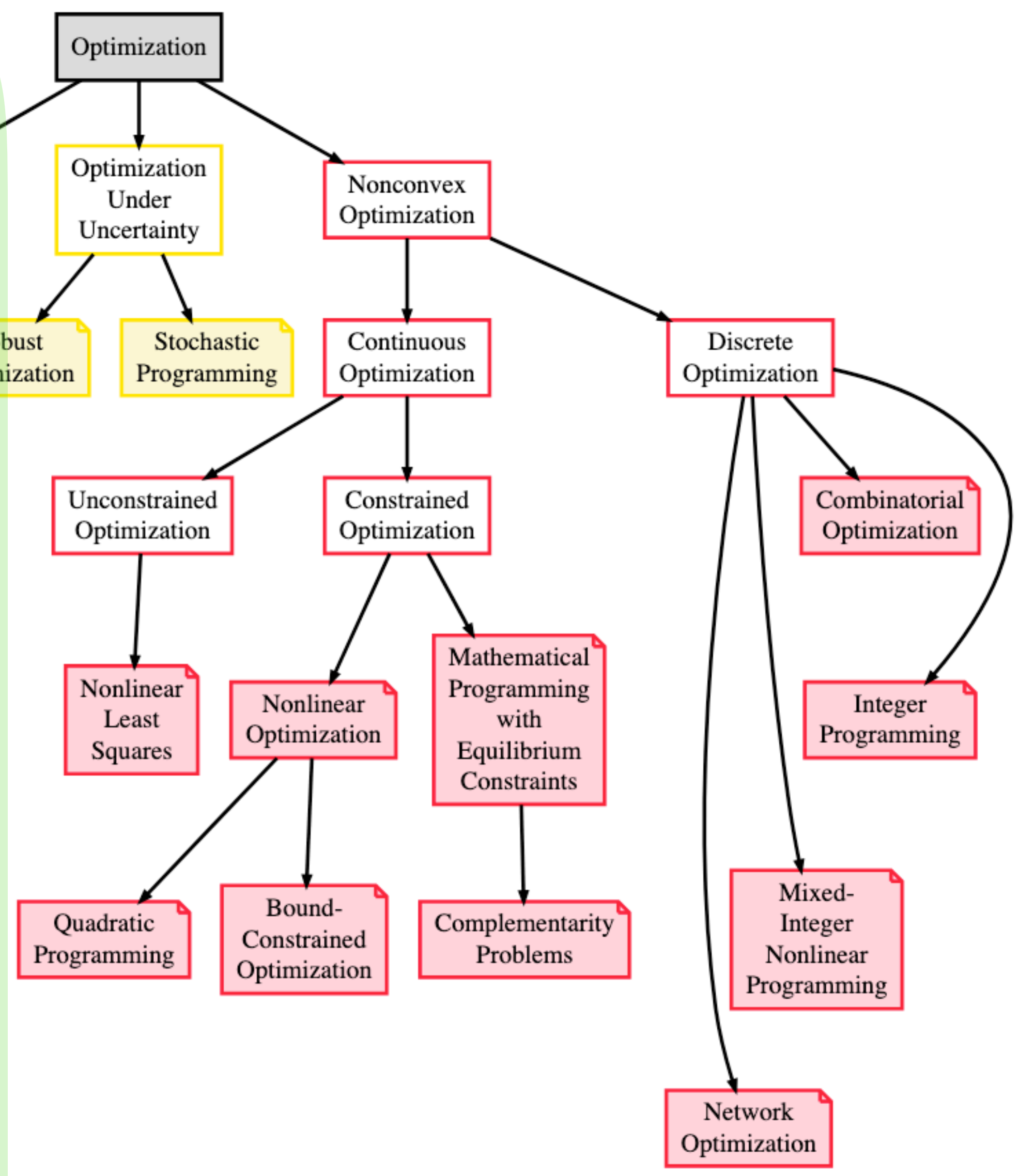
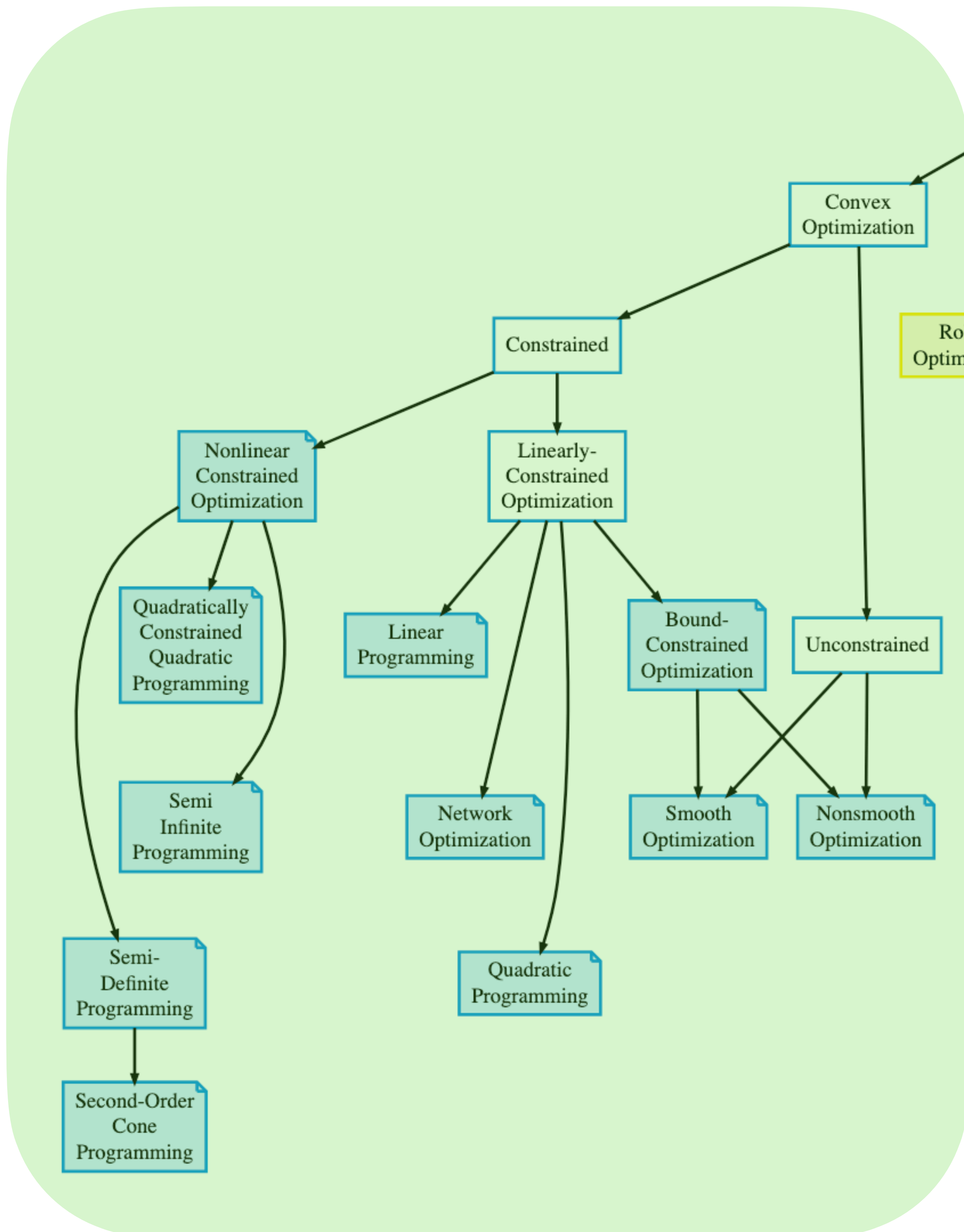


Convex



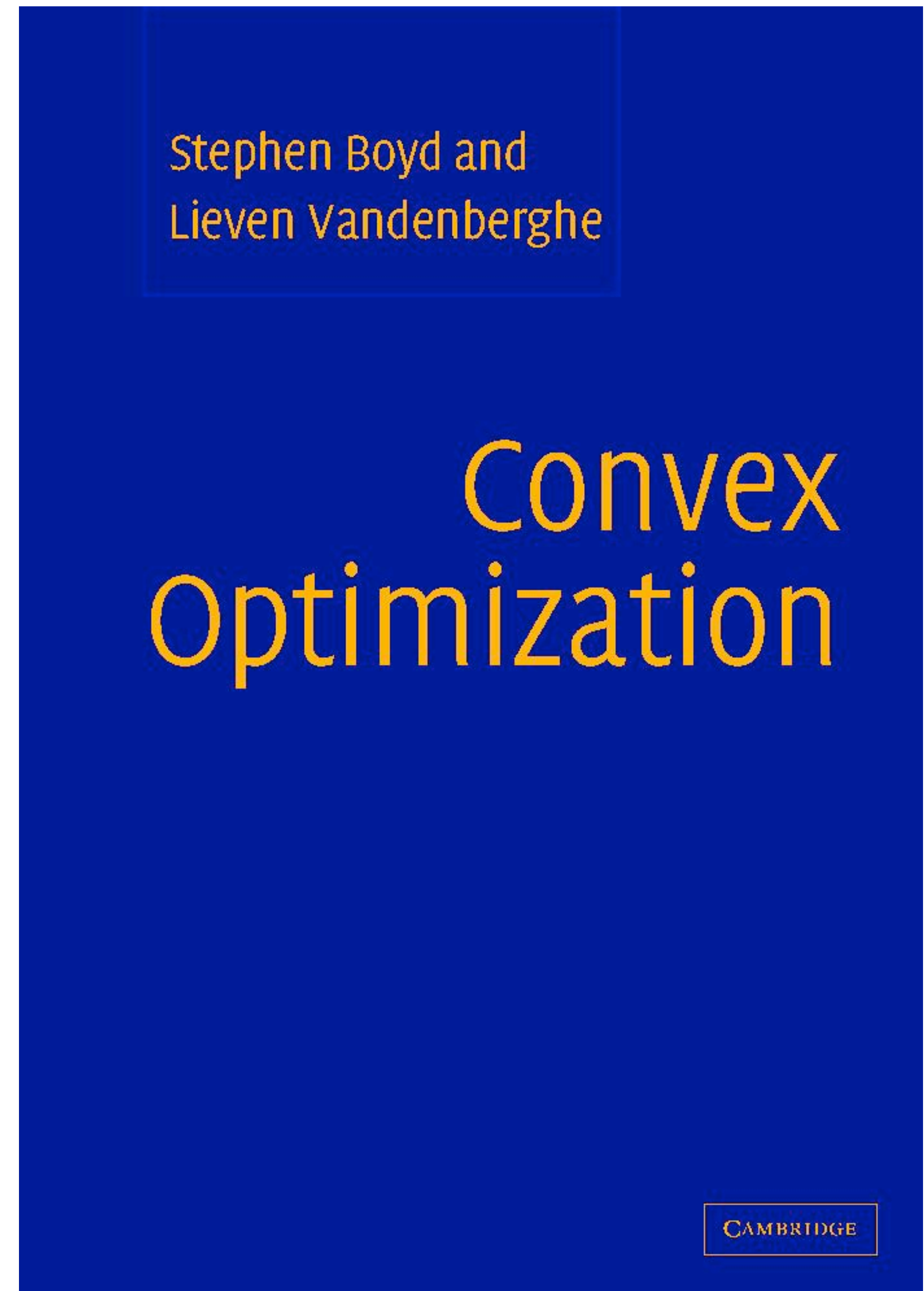
Non-convex





Convexity

- A whole field of optimization on its own
- Convex $\neq$ easy
- Advantages
 - (1) Enjoys strong convergence guarantees
 - (2) Local optimizer is global one



Convexity

Convex sets

- A set $\mathcal{X}$ is convex if $\forall x_1, x_2 \in \mathcal{X}$ and $\theta \in [0,1]$

$$\theta x_1 + (1 - \theta)x_2 \in \mathcal{X}$$

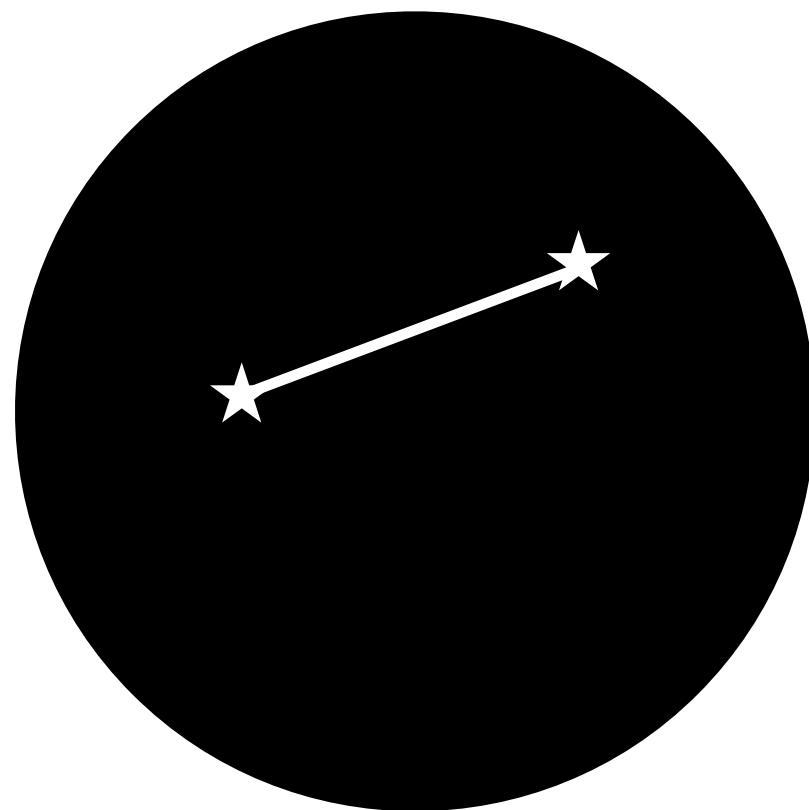
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$$\|x\|_2 \leq r_{\text{dist}}$$



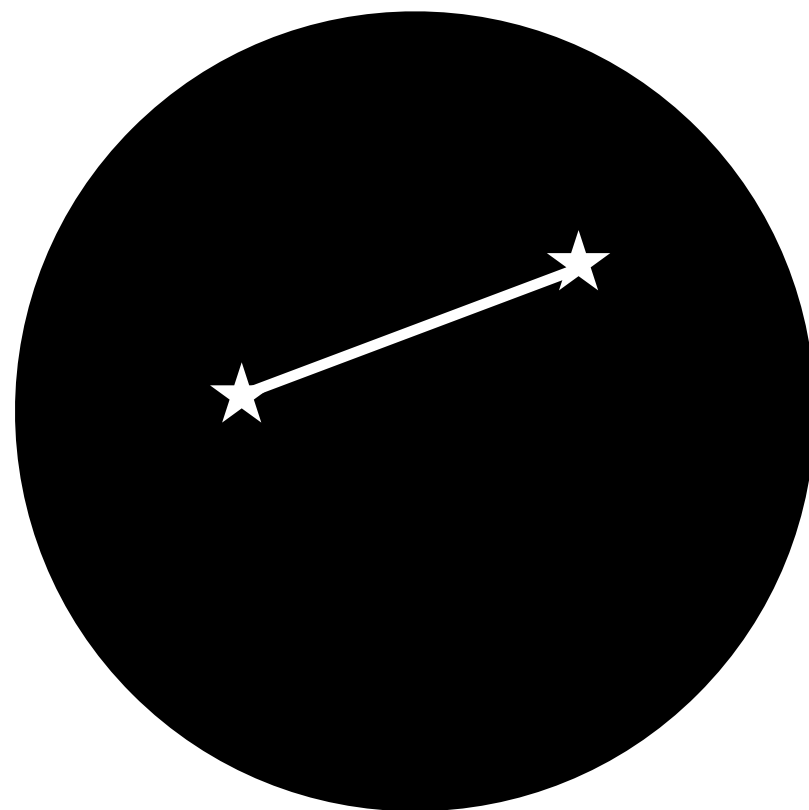
Convexity

Convex sets

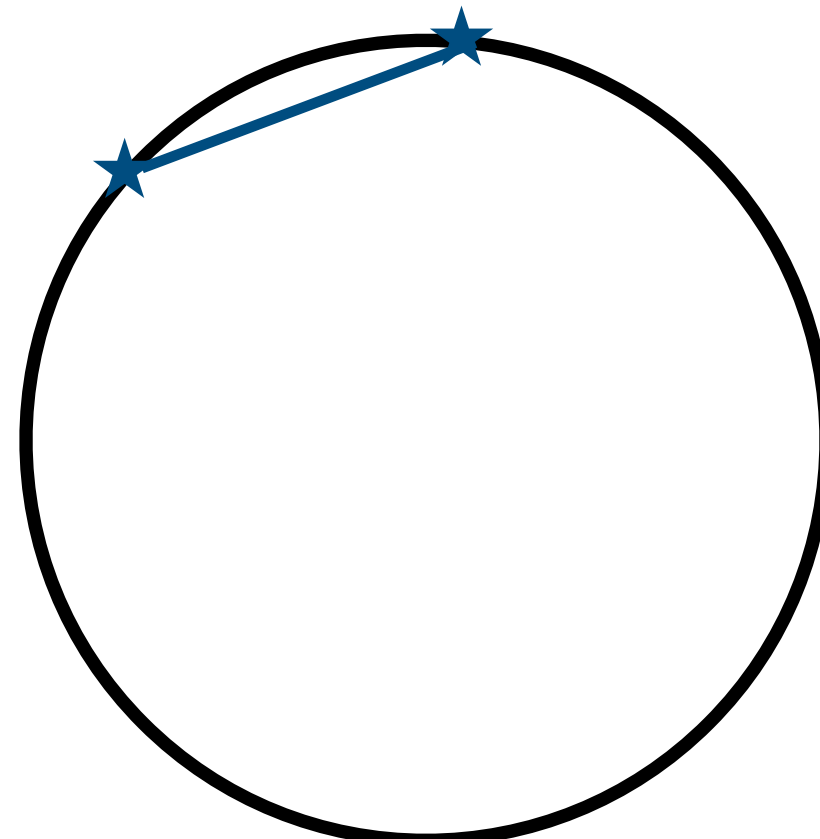
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$$\|x\|_2 = r_{\text{dist}}$$



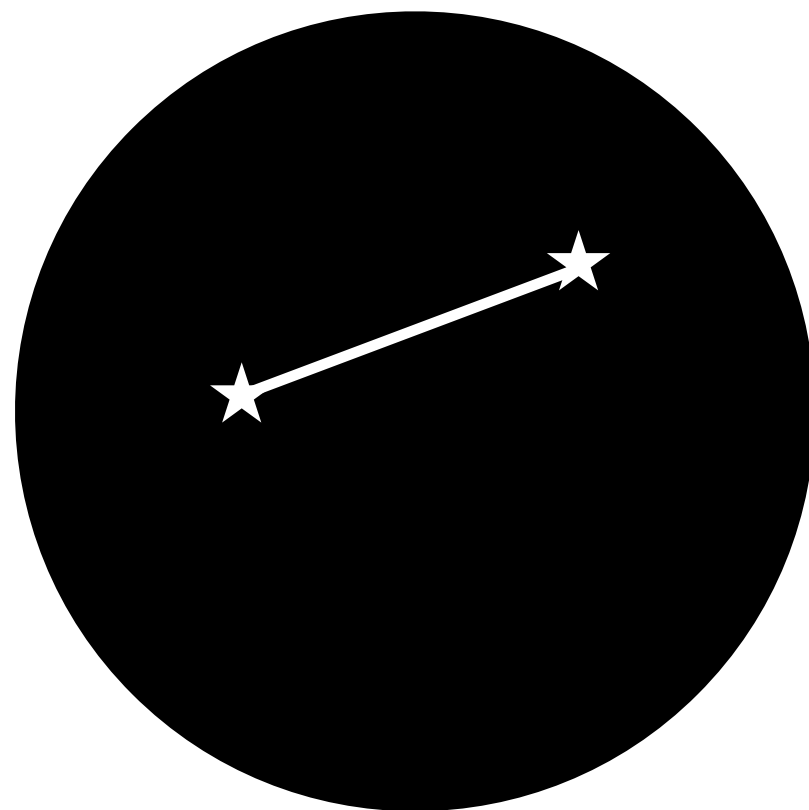
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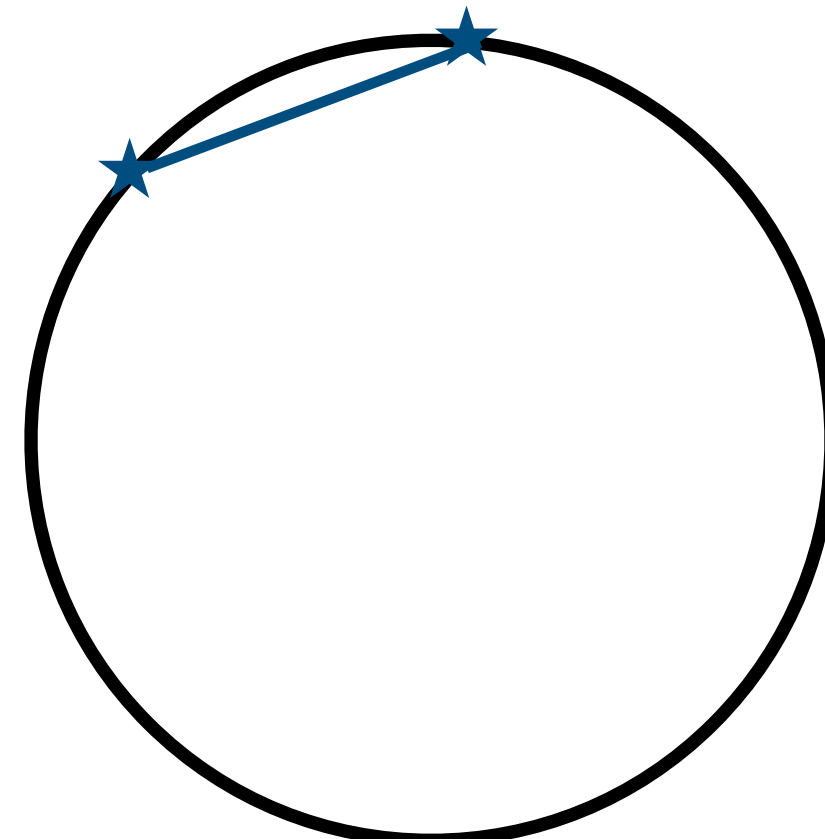
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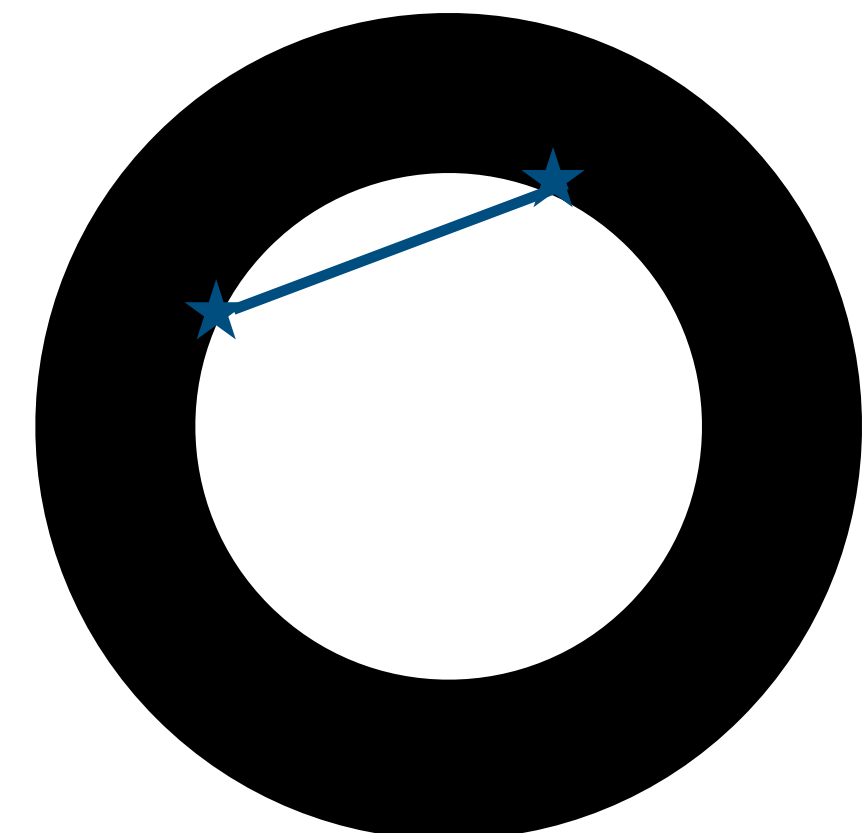
$$\|x\|_2 \leq r_{\text{dist}}$$



$$\|x\|_2 = r_{\text{dist}}$$



$$r_{\text{min}} \leq \|x\|_2 \leq r_{\text{max}}$$

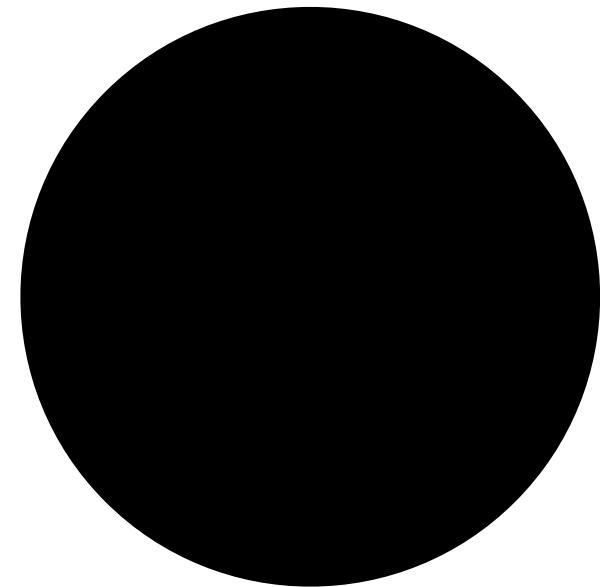


Convexity

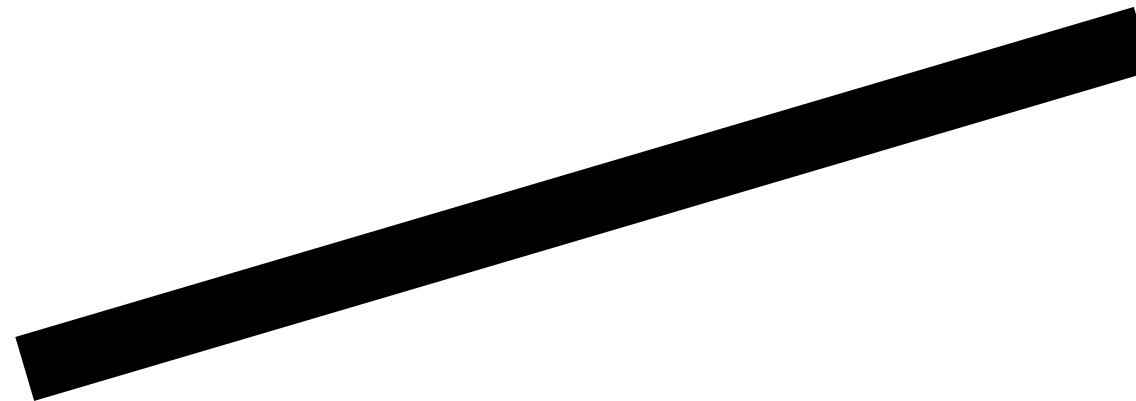
Convex sets

- The intersection of convex sets is convex

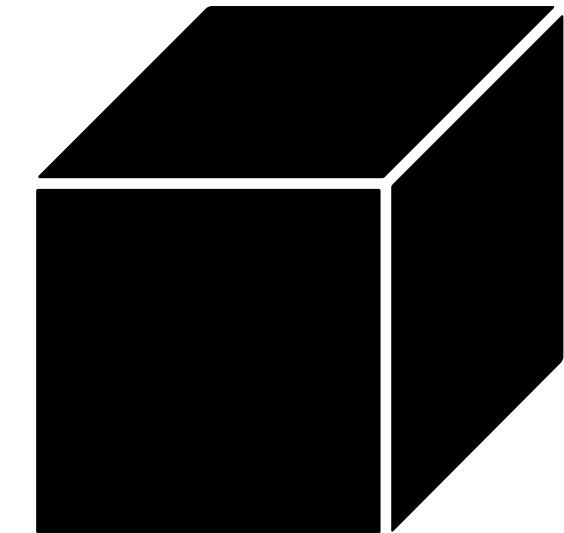
$$\|x\|_2 \leq r_{\text{dist}}$$



$$y = c^T x + b$$



$$\|u\|_1 \leq u_{\text{max}}$$



Convexity

Convex functions

- A function $f(x)$ is convex if $\forall x_1, x_2 \in \text{dom } f$ and $\theta \in [0, 1]$

$$f(\theta x_1 + (1 - \theta)x_2) \leq \theta f(x_1) + (1 - \theta)f(x_2)$$

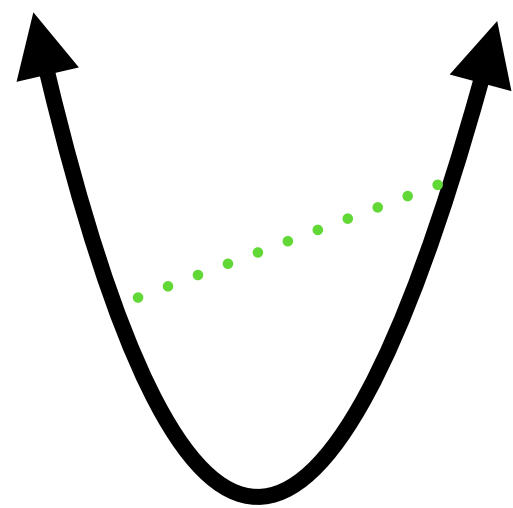
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$$f(x) = x^2$$



Convex

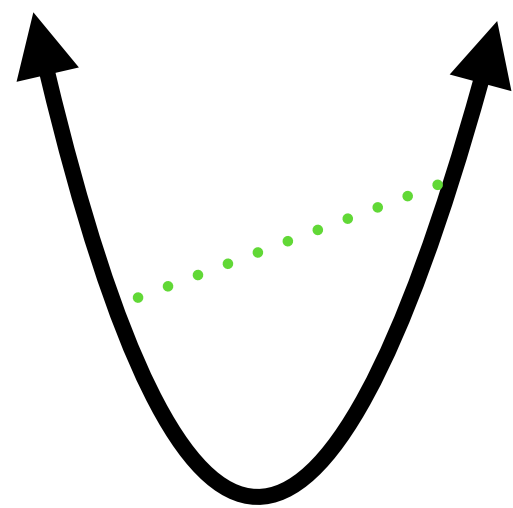
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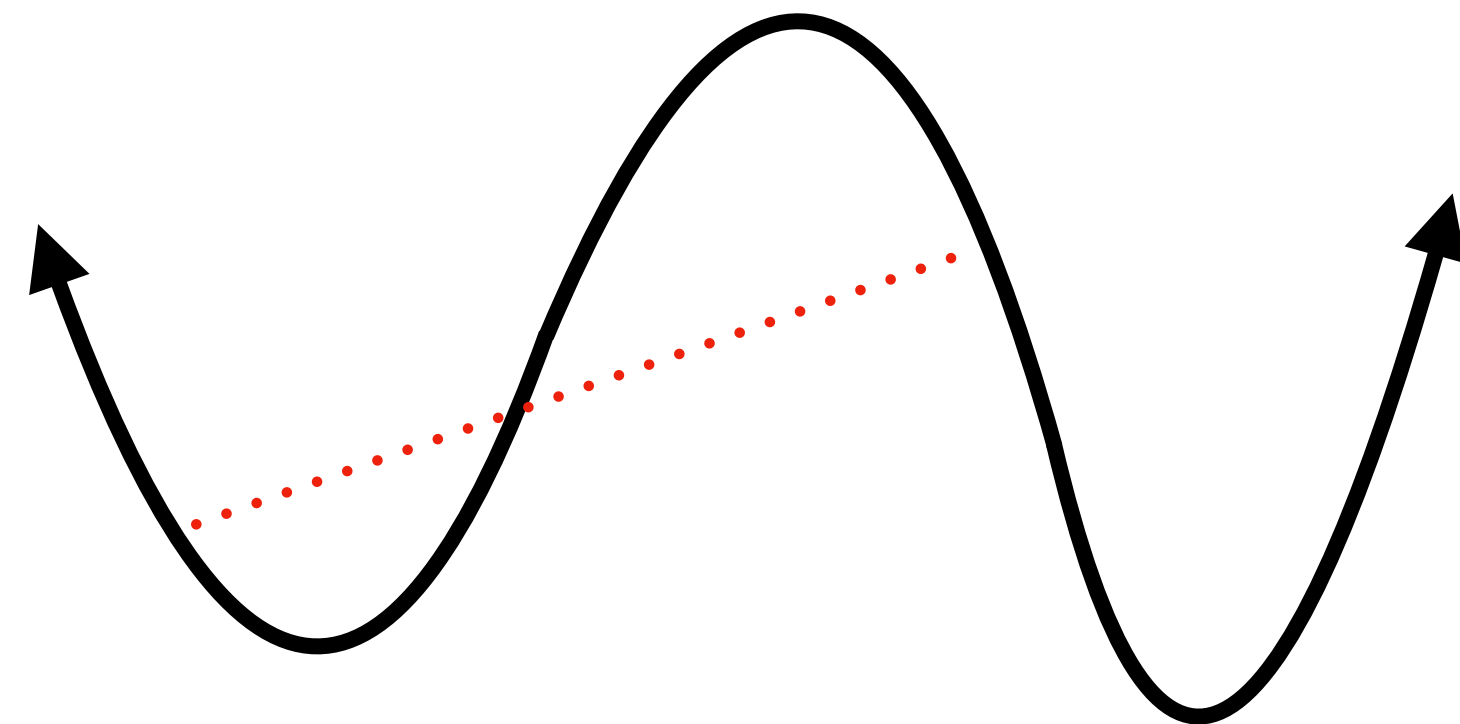
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$$f(x) = x^2$$



Convex

$$f(x) = x^4 - 2 * x^3 - 10x^2 + 7x - 17$$



Non-convex

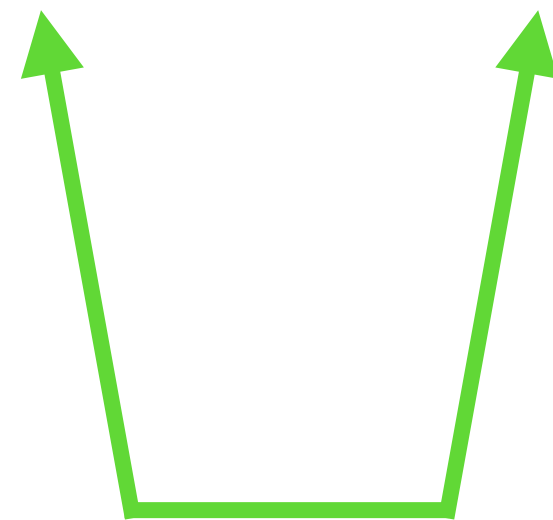
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$$f(\theta x_1 + (1 - \theta)x_2) \leq \theta f(x_1) + (1 - \theta)f(x_2)$$

- **Note:** f need not be smooth (i.e., $\nabla_x f(x)$ might not exist)



Convex



Non-convex

Convexity

Convex functions and sets

- The epigraph of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is:

$$\text{epi } f = \{ (x, t) \mid x \in \text{dom } f, f(x) \leq t \}$$

Convexity

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- f is a convex function if and only if $\text{epi } f$ is a convex set

$$f \text{ is convex function} \Leftrightarrow \text{epi } f \text{ is convex set}$$

Convexity

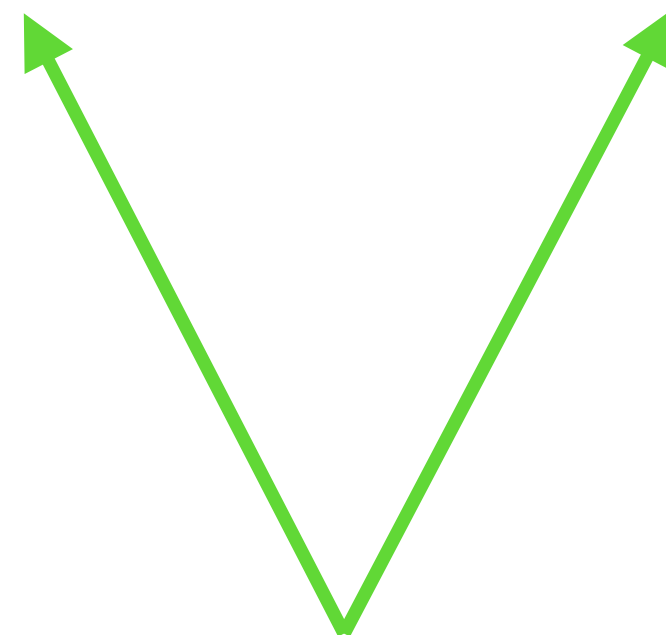
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Convexity

Why does this matter?

- A *local* solution to a convex problem is also the *global* solution
 - For smooth unconstrained optimization, if $f \in \mathcal{C}^2$ and convex:

$$x^* \text{ is global optimizer} \Leftrightarrow \nabla_x f(x^*) = 0$$

- For smooth constrained optimization, if $f, h, g \in \mathcal{C}^2$ and convex:

$$x^* \text{ is global optimizer} \Leftrightarrow \nabla_x \mathcal{L}(x^*, y^*, z^*) = 0$$

Convexity

Completeness guarantees

- A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called *self-concordant* if f is convex and:

$$f'''(x) \leq 2f''(x)^{\frac{3}{2}}$$

- Loosely, “third derivative doesn’t change too quickly”
- Interior point methods solve convex optimization problems in polynomial time

$$N = \mathcal{O}\left(\sqrt{n} \log\left(\frac{1}{\varepsilon}\right)\right)$$

Convexity

Interior-Point Polynomial
Algorithms in Convex
Programming

Yurii Nesterov
Arkadii Nemirovskii

siam Studies in
Applied Mathematics

THE INTERIOR-POINT REVOLUTION IN OPTIMIZATION: HISTORY, RECENT DEVELOPMENTS, AND LASTING CONSEQUENCES

MARGARET H. WRIGHT

1. OVERVIEW

REVOLUTION:

- (i) a sudden, radical, or complete change;*
- (ii) a fundamental change in political organization, especially the overthrow or renunciation of one government or ruler and the substitution of another*¹

It can be asserted with a straight face that the field of continuous optimization has undergone a revolution since 1984 in the sense of the first definition and that the second definition applies in a philosophical sense: Because the interior-point presence in optimization today is ubiquitous, it is easy to lose sight of the magnitude and depth of the shifts that have occurred during the past twenty years. Building on the implicit political metaphor of our title, successful revolutions eventually become the status quo.

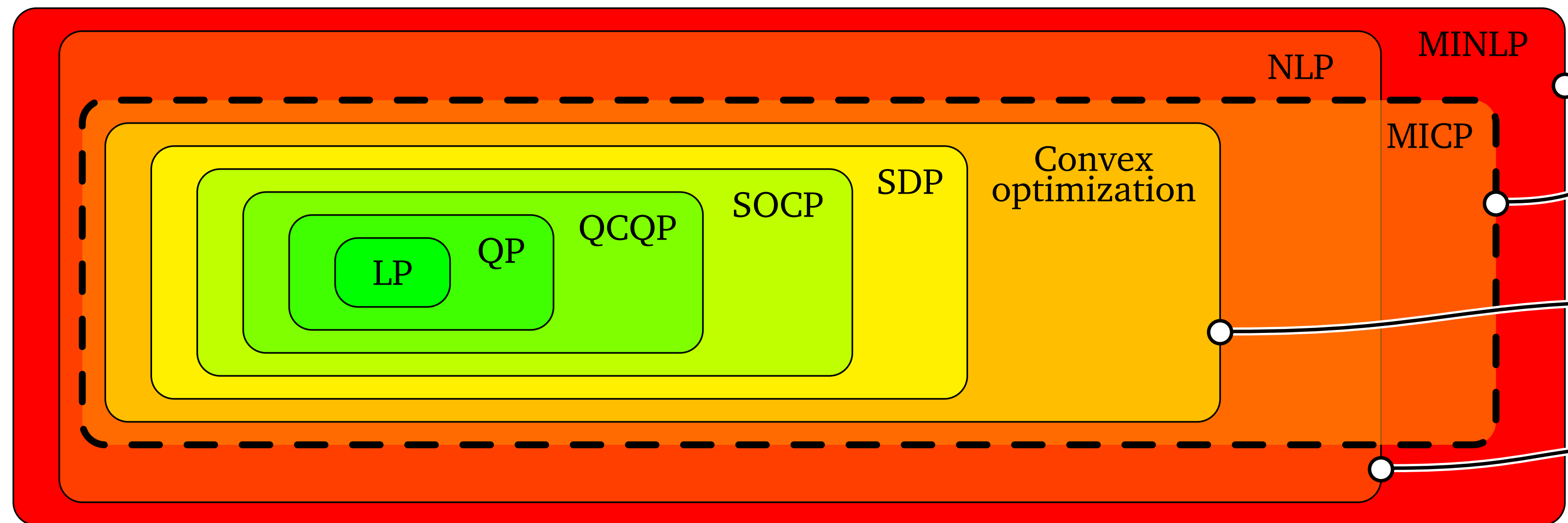
Convexity

In summary

Convexity

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- Convexity does not mean the problem is *easy*

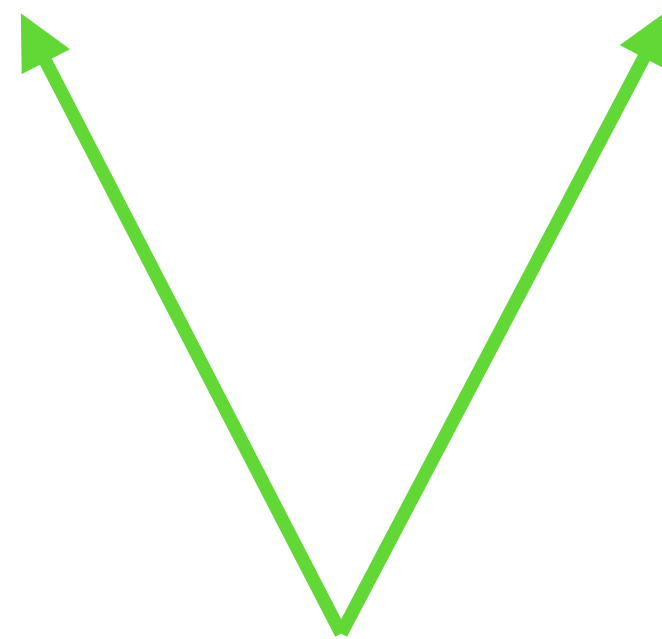


D. Malyuta, Y. Yu, P. Elango, and B. Acikmese, "Advances in trajectory optimization for space vehicle control," *Annual Reviews in Control*, vol. 52, pp. 282 — 315, 2021.

Convexity

In summary

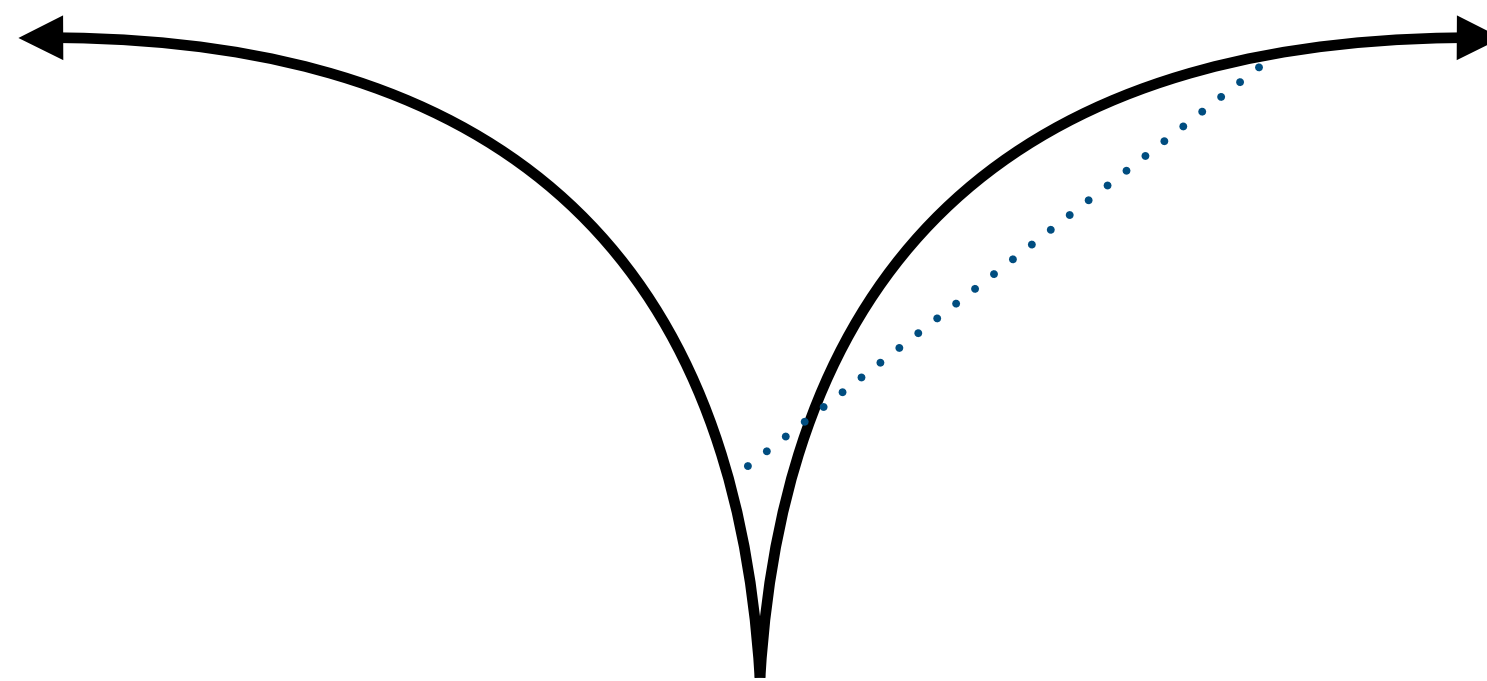
- Convexity does not mean the problem is *easy*
- Convex programs do not have to be smooth



Convexity

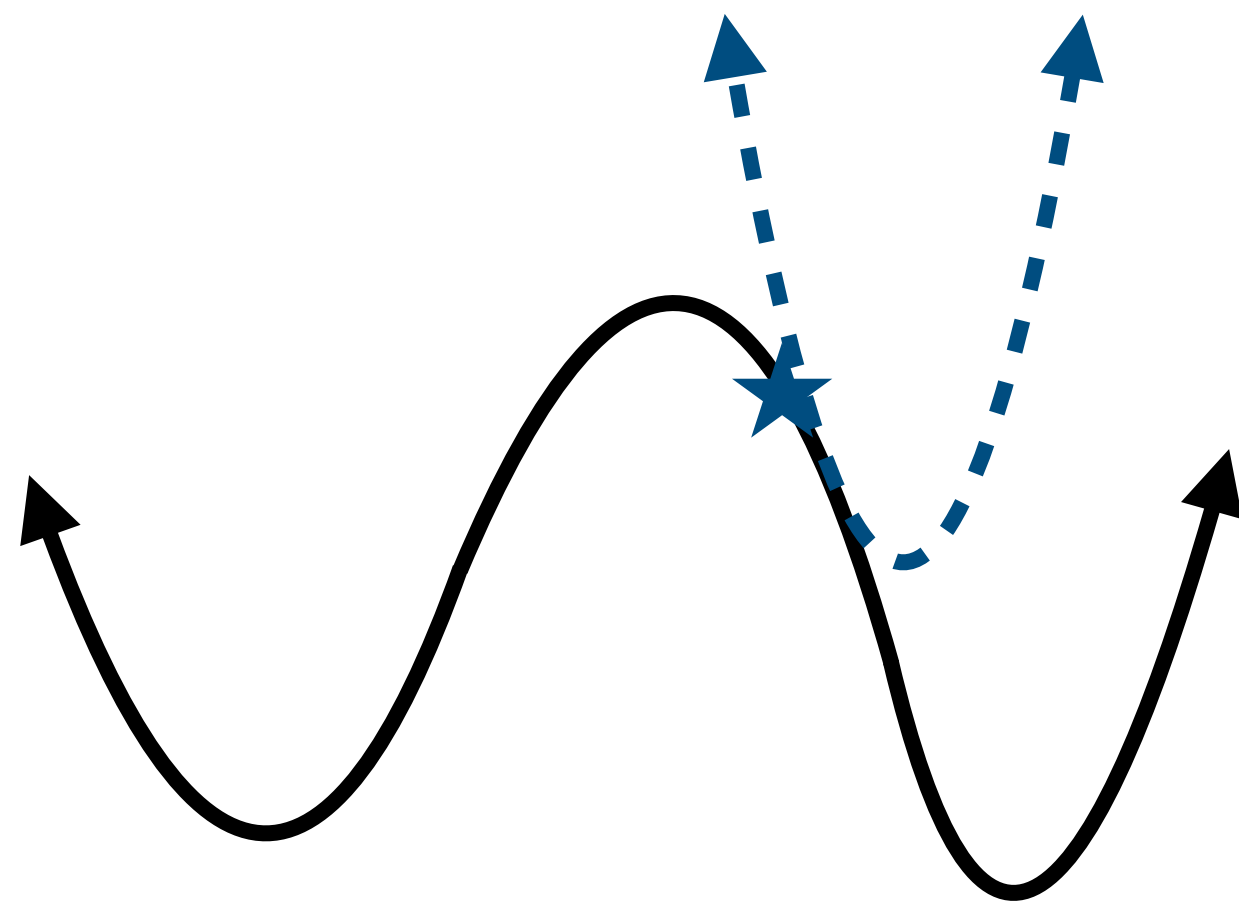
In summary

- Convexity does not mean the problem is *easy*
- Convex programs do not have to be smooth
- Convex programs are not the only ones with global optima



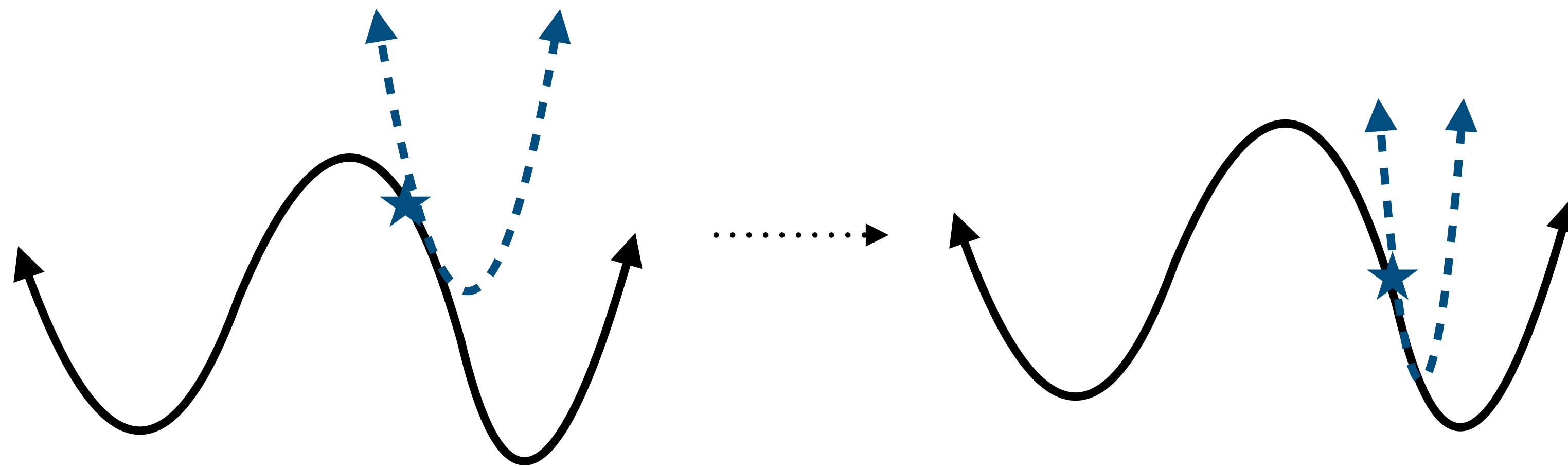
Relevance to trajectory design

- Most practical systems of interest are non-convex
- For *smooth* non-convex optimization problems, most approaches rely on solving a sequence of convex approximations to the problem



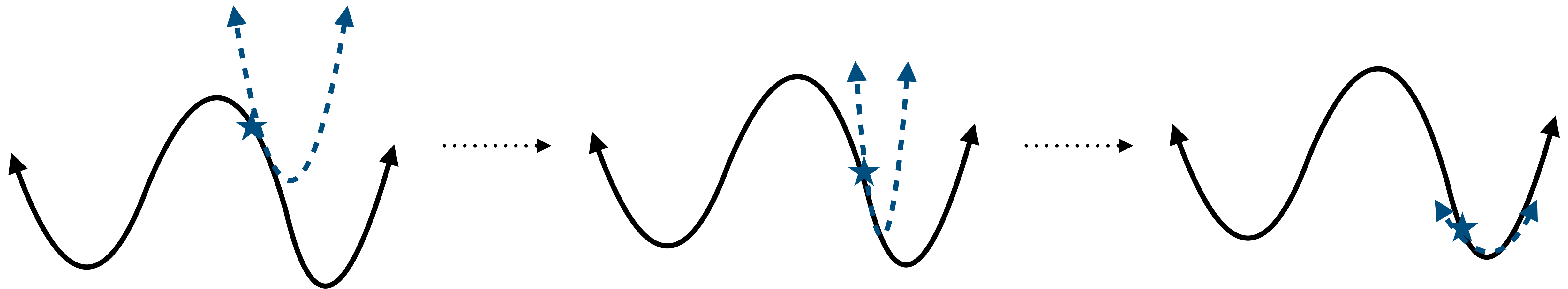
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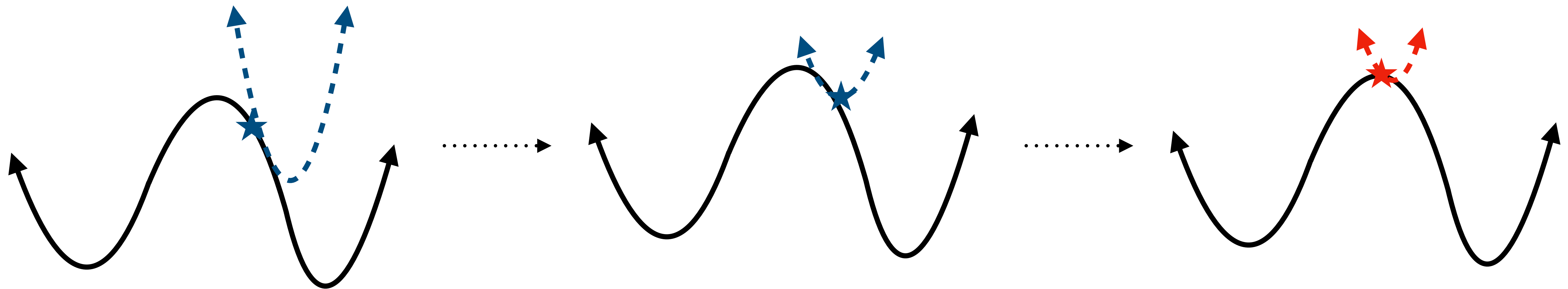
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Relevance to trajectory design

- Most practical systems of interest are non-convex
- For *smooth* non-convex optimization problems, most approaches rely on solving a sequence of convex approximations to the problem
- Caveat: it's easy to get stuck in local minima



Relevance to trajectory design

Sequential quadratic programming

$$\begin{array}{ll} \min_x & f(x) \\ \text{subj. to:} & h(x) = 0 \\ & g(x) \leq 0 \\ & f, h, g \in \mathcal{C}^2 \end{array}$$

Relevance to trajectory design

Sequential quadratic programming

$$\begin{array}{ll} \min_x & f(x) \\ \text{subj. to:} & h(x) = 0 \\ & g(x) \leq 0 \\ & f, h, g \in \mathcal{C}^2 \end{array}$$



1. Given $w^0 = (x^0, y^0, z^0)$

2. Construct Lagrangian using w

for $k \in [1, N_{\max}]$

3. Solve local QP approximation for Δw^k

$$\begin{array}{ll} \min_{\Delta x} & \frac{1}{2} \Delta x^T \nabla_{xx}^2 \mathcal{L}(\bar{x}) \Delta x + \nabla_x \mathcal{L}(\bar{x}) \Delta x \\ \text{subj. to:} & h(\bar{x}) + \nabla_x h(\bar{x}) \Delta x = 0 \\ & g(\bar{x}) + \nabla_x g(\bar{x}) \Delta x \leq 0 \end{array}$$

4. Line search to find α

$$5. w^k = w^{k-1} + \alpha \Delta w^k$$

6. Terminate when necessary conditions satisfied

Convexity of practical constraints

Convexity of practical constraints

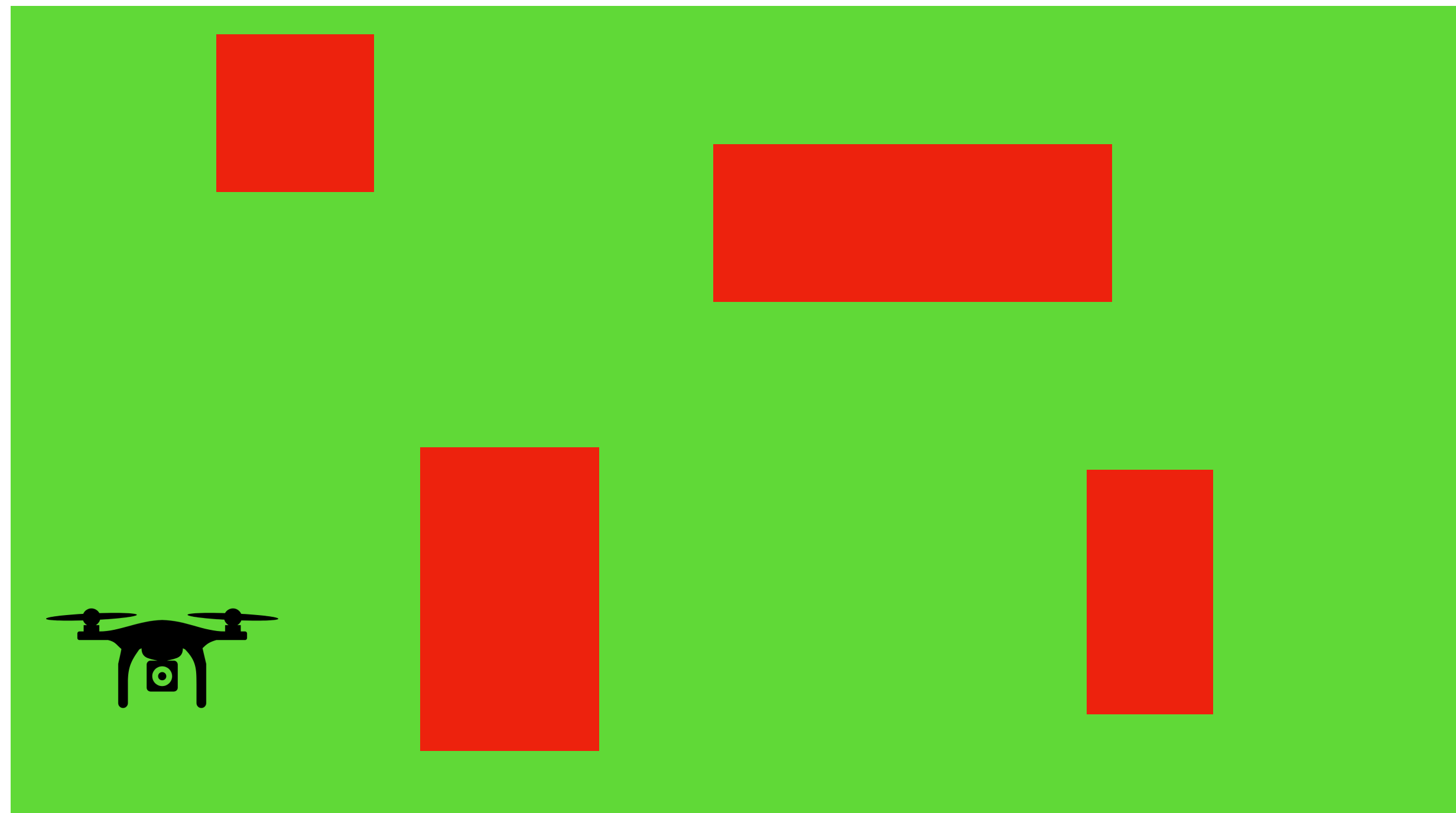
Collision avoidance



$$x_k \notin \mathcal{X}_{\text{obs}}$$

Convexity of practical constraints

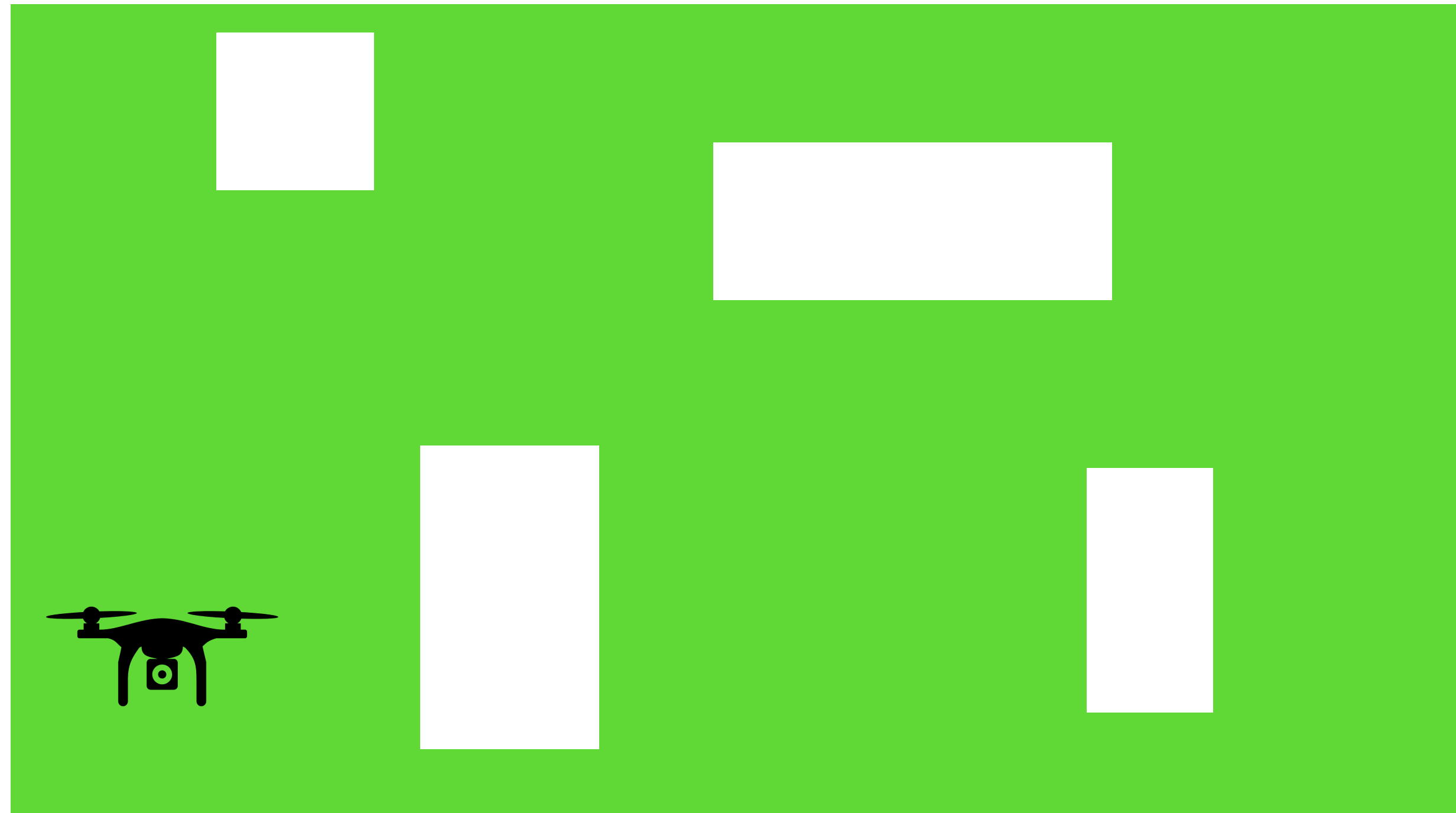
Collision avoidance



$$x_k \in \mathcal{X}_{\text{safe}}$$

Convexity of practical constraints

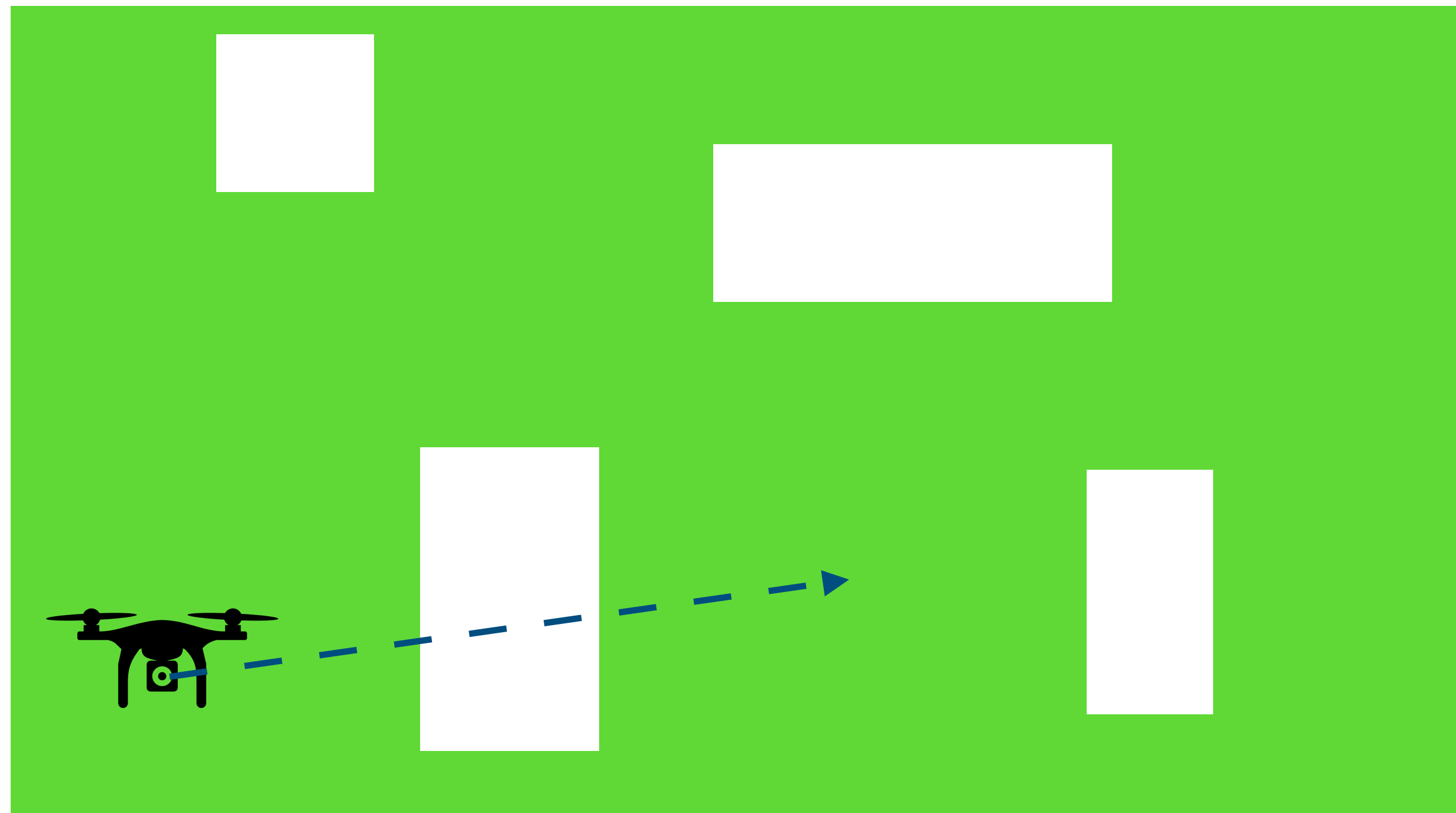
Collision avoidance



$$x_k \in \mathcal{X}_{\text{safe}}$$

Convexity of practical constraints

Collision avoidance



$$x_k \in \mathcal{X}_{\text{safe}} \quad \text{Non-convex}$$

Convexity of practical constraints

Dynamics constraints

- An equality constraint can be written as a two-sided inequality

$$x_{k+1} = f(x_k, u_k) \quad \longleftrightarrow \quad \begin{aligned} x_{k+1} &\leq f(x_k, u_k) \\ x_{k+1} &\geq f(x_k, u_k) \end{aligned}$$

Convexity of practical constraints

Dynamics constraints

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- For both a function $f(x_k, u_k)$ and its negative $-f(x_k, u_k)$ to be convex with respect to x_k and u_k , $f(\cdot)$ must be an affine function

$$x_{k+1} = Ax_k + Bu_k + c$$

Convexity of practical constraints

Dynamics constraints

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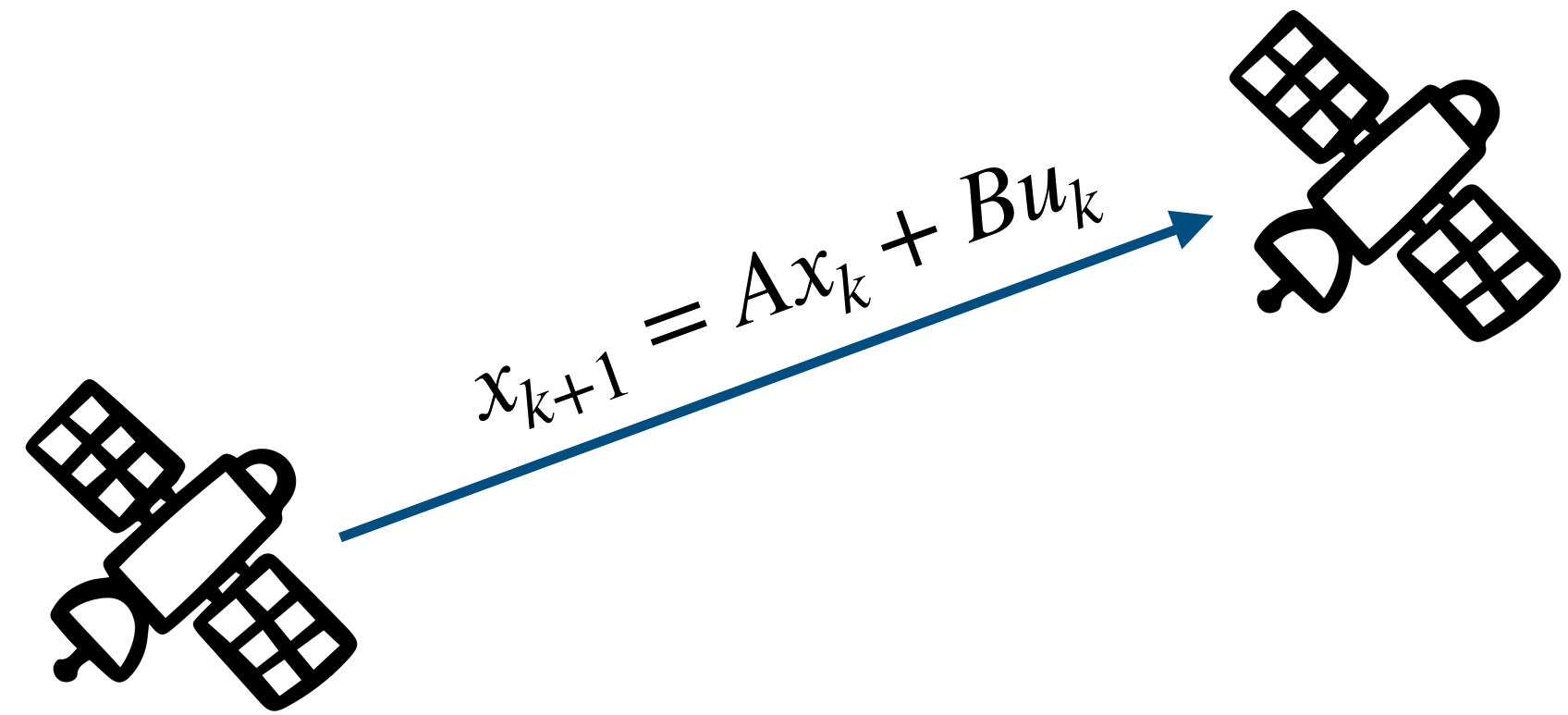
- Any system with nonlinear dynamics is non-convex

Convexity of practical constraints

Dynamics constraints

Spacecraft double integrator (convex)

$$x_{k+1} = \begin{pmatrix} I & \Delta t I \\ 0 & I \end{pmatrix} x_k + \begin{pmatrix} \Delta t^2 I \\ \Delta t I \end{pmatrix} u_k$$

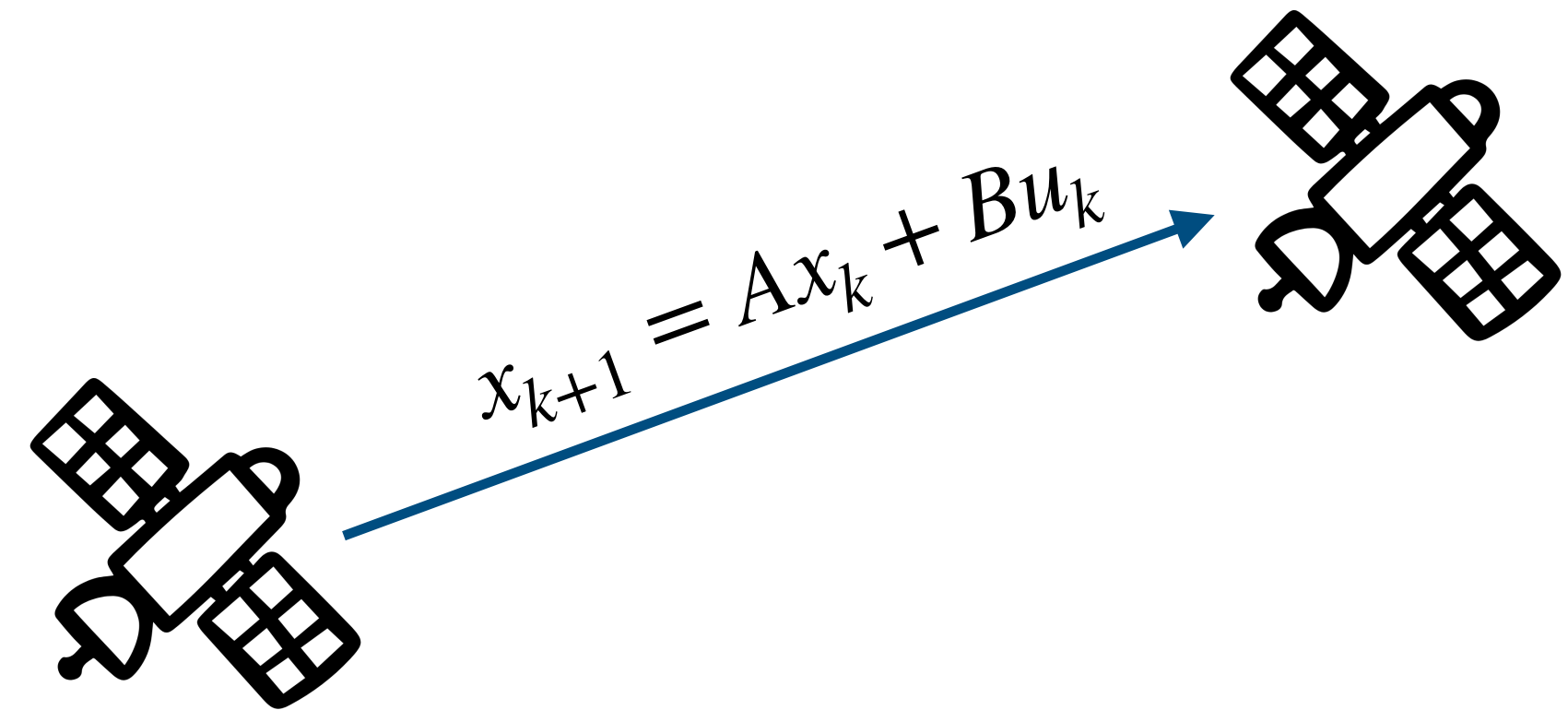


Convexity of practical constraints

Dynamics constraints

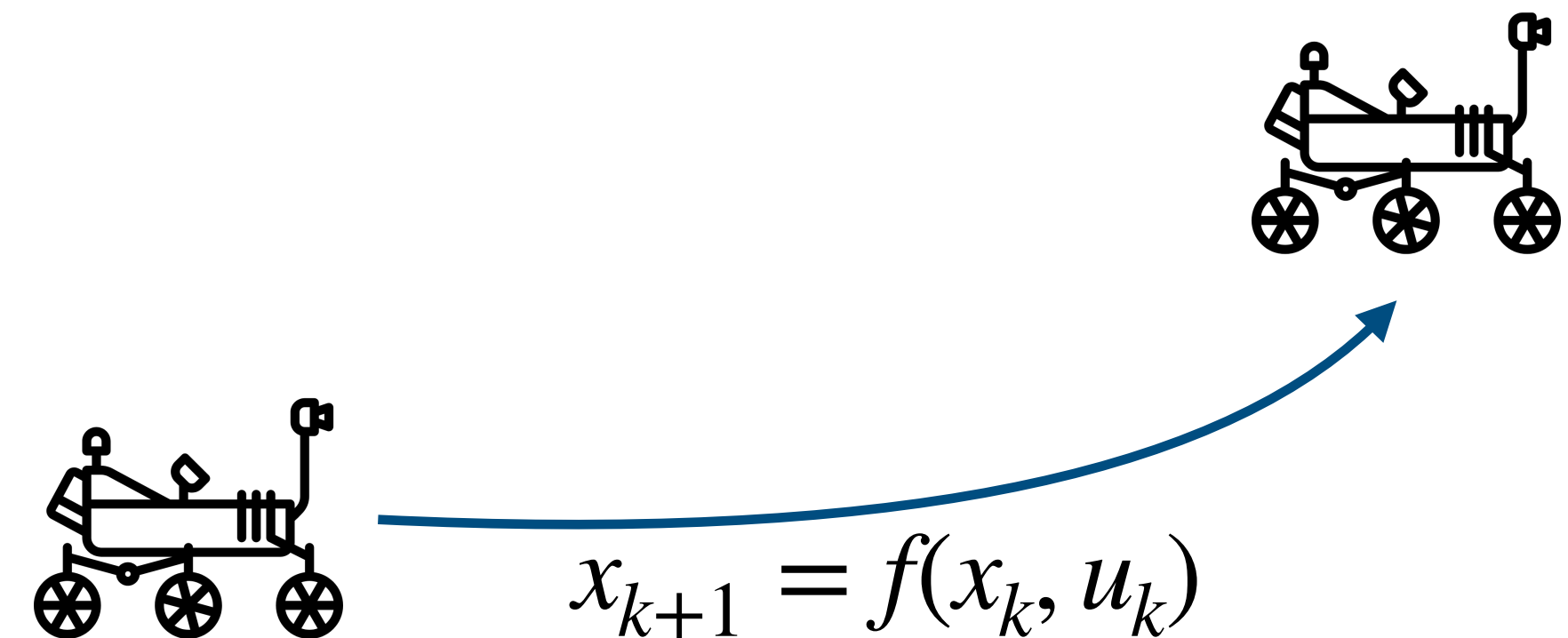
Spacecraft double integrator (convex)

$$x_{k+1} = \begin{pmatrix} I & \Delta t I \\ 0 & I \end{pmatrix} x_k + \begin{pmatrix} \Delta t^2 I \\ \Delta t I \end{pmatrix} u_k$$



Mars rover (non-convex)

$$x_{k+1} = x_k + \Delta t \begin{pmatrix} v_k \cos \theta_k \\ v_k \sin \theta_k \\ \omega_k \end{pmatrix}$$



Optimal control as nonlinear programs

How to convert a trajectory generation problem into standard optimization form?

Solve for trajectory $(x_{0:N}, u_{0:N})$

$$\min_{x_{0:N}, u_{0:N}} g_T(x_N) + \sum_{k=0}^{N-1} g_k(x_k, u_k)$$

subject to:

$$x_0 = x_{\text{init}}$$

$$x_{k+1} = f(x_k, u_k)$$

$$x_k \in \mathcal{X}_{\text{safe}}$$

$$u_k \in \mathcal{U}$$

Optimal control as nonlinear programs

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subject to:

$$x_0 = x_{\text{init}}$$

$$x_{k+1} = f(x_k, u_k)$$

$$x_k \in \mathcal{X}_{\text{safe}}$$

$$u_k \in \mathcal{U}$$

1. Convert to standard form with inequality constraints written as $g_i(x_k, u_k) \leq 0$ and equality constraints as $h_i(x_{k+1}, x_k, u_k) = 0$

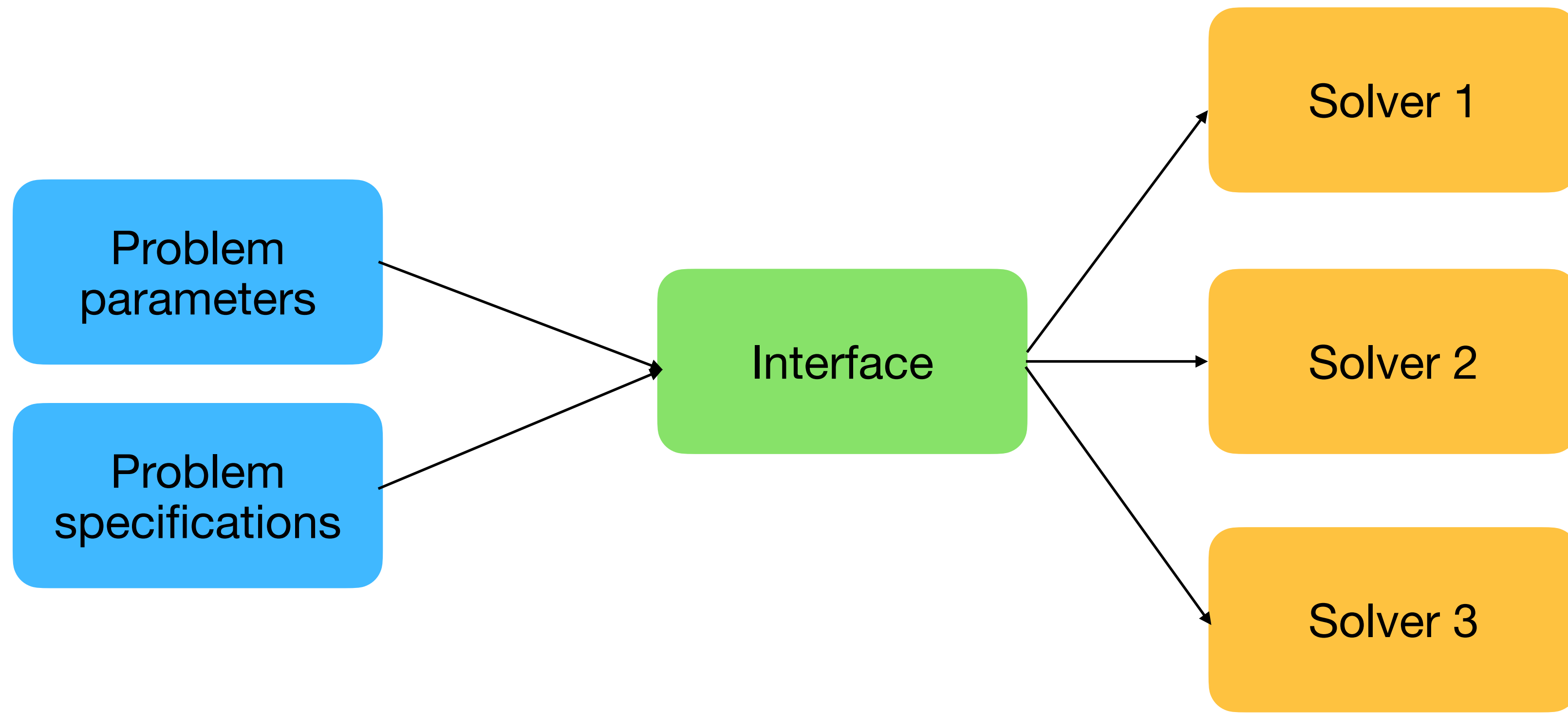
2. Provide gradient and Hessian for each constraint and cost

Converting an optimal control problem into standard form can be cumbersome!

Off-the-shelf trajectory optimization

Interface vs. solver

- An **interface** allows for solving the same optimization problem with different backend solvers
- Allows for rapid prototyping and testing



Off-the-shelf trajectory optimization

Interface vs. solver

```
1 X = cp.Variable(shape=(nx, N+1))
2 U = cp.Variable(shape=(nu, N))
3
4 cost = 0.0
5 constraints = []
6
7 # Initial condition for system
8 constraints += [X[:,0] == x_init]
9
10 for cp_idx in range(N):
11     # Add stage-wise cost
12     cost += cp.quad_form(X[:, cp_idx+1] - x_goal, Q) + cp.quad_form(U[:, cp_idx], R)
13
14     # Dynamics constraint
15     constraints += [X[:, cp_idx+1] == Ak @ X[:,cp_idx] + Bk @ U[:,cp_idx]]
16
17     # State upper and lower bounds
18     constraints += [X[:, cp_idx+1] <= x_max]
19     constraints += [X[:, cp_idx+1] >= x_min]
20
21     # Control upper and lower bounds
22     constraints += [U[:, cp_idx] <= u_max]
23     constraints += [U[:, cp_idx] >= u_min]
24
25 prob = cp.Problem(cp.Minimize(cost), constraints)
26 prob.solve()
27 print(prob.value)
```

CVXPY

Off-the-shelf trajectory optimization

Engineering considerations

- Linear algebra routines
- Problem scaling
- Presolve techniques
- Infeasibility detection
- Parameter tuning

Solver interfaces

Python



Julia



jump-dev/**Convex.jl**

A Julia package for disciplined convex programming



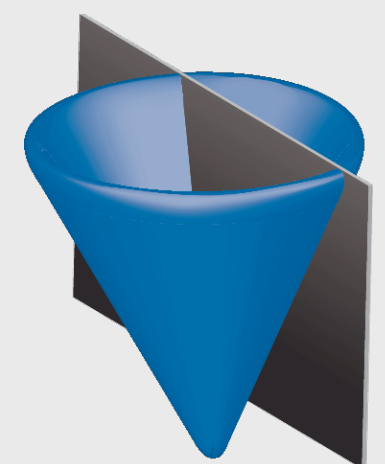
C++



MATLAB

yalmip/**YALMIP**

MATLAB toolbox for optimization modeling



CVX

Convex solvers

Linear programs: GLPK, SCIP

Quadratic programs: OSQP, HPIPM, QPOASES

Second-order cone programs: SCS, ECOS,
Clarabel

Semidefinite programs: SEDUMI, SDPT3

General nonlinear programs: IPOPT, SNOPT,
KNITRO

Mixed integer linear programs: CPLEX, KNITRO,
MOSEK, Gurobi

Mixed integer quadratic programs: MOSEK, Gurobi

Mixed integer nonlinear programming

NLLS solvers

Popular for SLAM

- Many problems in SLAM yield nonlinear least squares (NLLS) problems

$$\min_x \|f(x) - y\|_2^2$$

- Constraints are managed by “penalizing” violations
- Weaker guarantees and recent work in SLAM has sought to connect convex optimization with such problems
- Solvers: g2o, GTSAM, ceres, symforce, theseus

References

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- D. Malyuta, T. P. Reynolds, M. Szmuk, T. Lew, R. Bonalli, M. Pavone, and B. Acikmese, “Convex Optimization for Trajectory Generation,” *IEEE Control Systems Magazine*, vol. 42, no. 5, pp. 40 — 113, 2022.